



DEDICATION

I dedicate this humble work to my entire family and to all those who supported me closely and from afar throughout my academic journey.

Very special dedications to:

- My father, Mr. SORHO YETIENA, for his benevolent guidance, wisdom and precious advice;
- My mother, Mrs. SORHO SANABA, for her unconditional love, constant prayers and unfailing support;
- My sister, SORHO ESLIE, for her permanent encouragement and comforting dynamism.

To all those who helped make me the person I am today, please find here the expression of my deep gratitude.

ACKNOWLEDGEMENTS

I would like to express my sincere gratitude to Mr. Kologo Harouna, Director of Soutarah Group, as well as to my internship supervisor, for their warm welcome, guidance, advice and availability throughout my internship. Also, this report was made possible thanks to the involvement of several people, and I would like to express my thanks to:

- My parents, Mr. Yetiena SORHO and Mrs. Sanaba COULIBALY, for all their sacrifices, and their invaluable financial and moral support since the first day of my studies;
- The Republic of Côte d'Ivoire, for all the university infrastructure and the framework of excellence offered to student youth;
- The Félix Houphouët-Boigny National Polytechnic Institute (INP-HB) of Yamoussoukro, for opening its doors to me and providing me with elite technical education;
- Dr. Moussa DIABY Abdoul Kader, Managing Director of INP-HB, for his inspiring leadership and constant commitment to the institute's prestige;
- Dr. Adama OUATTARA, Director of the Higher School of Industry (ESI), for the reforms and rigor instilled within our school;
- Mr. Siriky KONE, Deputy Director of Studies at ESI, for his attentive listening, availability and wise advice;
- Mr. Louagbeu Loua KPO, Head of the Information and Communication Sciences and Technologies (STIC) Teaching Unit, for the exceptional quality of the training program and his strategic guidance;
- The entire teaching and administrative staff of ESI and the STIC department for their dedication and passionate transmission of knowledge;
- The General Management and all staff of the company SOUTARAH GROUP, for their warm welcome, daily technical support and the trust granted during this professional immersion internship.

FOREWORD

The Félix HOUPHOUËT-BOIGNY National Polytechnic Institute (INP-HB) of Yamoussoukro was created by decree no. 96-678 of September 4, 1996, amended by decree no. 2016-747 of September 27, 2016. INP-HB results from the merger of four major engineering schools, which are:

- INSET : National Higher Institute of Technical Education;
- ENSTP : National Higher School of Public Works;
- ENSA : National Higher School of Agronomy;
- IAB : Agricultural Institute of Bouaké.

Today, INP-HB includes eleven (11) major schools responsible for the qualification and training of students. These include, among others:

- ESI : Higher School of Industry;
- ESMG : Higher School of Mines and Geology;
- ESA : Higher School of Agronomy;
- ESCAE : Higher School of Commerce and Business Administration;
- ESTP : Higher School of Public Works;
- EFSPC : School of Specialized Training and Executive Development;
- EDPSAPT : Doctoral School of Agronomic Sciences and Transformation Processes;
- ESCPE : Higher School of Petroleum Chemistry and Energy;
- EPGE : Preparatory School for Grandes Écoles;
- EDPSTI : Doctoral School of Engineering Sciences and Technologies;
- ESAS : Higher School of Aeronautics and Space.

INP-HB is a hallmark of reform that, as part of student development, offers programs aimed at training senior technicians, engineering technologists, and design engineers in the fields of industry, aeronautics, commerce, administration, civil engineering, public works, agronomy, mining and geology. In addition, INP-HB is also involved in research activities to contribute to the advancement of knowledge and technologies in these areas.

As part of this improvement process, the leadership of ESI implemented a reform in 2016 requiring its students to carry out immersion internships in the first year, application in-



ternships in the second year, and graduation professional internships in the third year so that they can apply the knowledge acquired throughout their academic journey.

Accordingly, I completed my internship within the company Soutarah Group. This 2-month internship took place from August 3, 2026 to October 3, 2026. This report recounts the activities we carried out during this period.

TABLE OF CONTENTS

DEDICATION	- 1 -
ACKNOWLEDGEMENTS.....	- 2 -
FOREWORD	- 3 -
TABLE OF CONTENTS	- 5 -
LIST OF FIGURES.....	- 6 -
LIST OF TABLES.....	- 7 -
LIST OF ABBREVIATIONS	- 8 -
ABSTRACT	- 9 -
GENERAL INTRODUCTION	- 10 -
PART I : PROJECT FRAMEWORK AND CONTEXT	- 11 -
Chapter 1 : Overview of the Host Organization	- 12 -
Chapter 2 : Project Presentation and Planning	- 15 -
PART II : CONCEPTUAL STUDY.....	- 19 -
Chapter 3 : Choice of Analysis Methodology.....	- 20 -
PART III : TECHNICAL STUDY.....	- 37 -
Chapter 4 : System Design and Technological Choices	- 38 -
PART IV : IMPLEMENTATION AND SYSTEM DEPLOYMENT.....	- 44 -
Chapter 5 : Development of the Web Platform and Mobile Application.....	- 45 -
Chapter 6 : Implementation	- 54 -
GENERAL CONCLUSION	- 57 -
BIBLIOGRAPHY	- 58 -
WEBOGRAPHY	- 59 -
TABLE OF CONTENTS	- 60 -

LIST OF FIGURES

Figure 1 : GANTT Diagram	- 18 -
Figure 2 : Global Use Case Diagram (Web & Mobile)	- 23 -
Figure 3 : Activity Diagram : Customer Account Creation and Management	- 27 -
Figure 4 : Activity Diagram : Vehicle Reservation	- 29 -
Figure 5 : Activity Diagram : Product Purchase	- 31 -
Figure 6 : Sequence Diagram : Secure Online Payment	- 32 -
Figure 7 : Sequence Diagram : Vehicle Reservation	- 33 -
Figure 8 : Sequence Diagram : Product Purchase	- 34 -
Figure 9 : Sequence Diagram : User Registration	- 35 -
Figure 10 : System Class Diagram	- 36 -
Figure 11 : Visual Studio Code Editor Logo	- 38 -
Figure 12 : Figma Logo	- 39 -
Figure 13 : React.js Framework Logo	- 39 -
Figure 14 : React Native & Expo mobile framework logo	- 39 -
Figure 15 : Node.js Logo	- 40 -
Figure 16 : Brevo Logo.....	- 40 -
Figure 17 : Genius Pay Logo	- 40 -
Figure 18 : Hostinger Logo.....	- 41 -
Figure 19 : General System Architecture Diagram	- 43 -
Figure 20 : Structure and Relational Schema of MySQL Database	- 45 -
Figure 21 : Web Interface : Home Page.....	- 48 -
Figure 22 : Web Interface : Login Page and Authentication Form	- 48 -
Figure 23 : Web Interface : Product Catalog	- 49 -
Figure 24 : Web Interface : Vehicle Reservation Module.....	- 49 -
Figure 25 : Web Interface : Multi-Item Cart and PDF Quotation	- 50 -
Figure 26 : Web Interface : Customer Portal and Request History	- 50 -
Figure 27 : Mobile Application : Home Screen and Detailed Vehicle Sheet	- 51 -
Figure 28 : Mobile Application : Mobile Booking Cart	- 52 -
Figure 29 : Administrative Interface : Dashboard	- 52 -
Figure 30 : Administrative Interface : Fleet and Reservation Management	- 53 -



LIST OF TABLES

Table 1 : Forecast Planning and Chronogram of Tasks	- 17 -
Table 2 : Comparative Study between MERISE and UP/UML Methodologies	- 21 -
Table 3 : Comparative Analysis of Database Management Systems	- 41 -
Table 4 : Acceptance Test Book and Functional Validation (Web & Mobile)	- 54 -
Table 5 : Estimated Financial Summary of Infrastructure and Annual Maintenance Costs	- 55 -

LIST OF ABBREVIATIONS

API	Application Programming Interface
BTP	Civil Engineering and Construction
CSS	Cascading Style Sheets
ESI	Higher School of Industry (INP-HB)
HTML	HyperText Markup Language
HTTP / HTTPS	HyperText Transfer Protocol (Secure)
IA	Artificial Intelligence
IDE	Integrated Development Environment
INP-HB	Félix Houphouët-Boigny National Polytechnic Institute
JSON	JavaScript Object Notation
JWT	JSON Web Token
LLM	Large Language Model
MCD	Conceptual Data Model
MLD	Logical Data Model
ORM	Object-Relational Mapping
PDF	Portable Document Format
PU	Unified Process
REST	Representational State Transfer
SARL	Limited Liability Company
SGBD	Database Management System
SGBDR	Relational Database Management System
SQL	Structured Query Language
STIC	Information and Communication Sciences and Technologies
UI / UX	User Interface / User Experience
UML	Unified Modeling Language

ABSTRACT

This application project focuses on the design and development of an overarching digital platform for managing the services of SOUTARAH GROUP, incorporating a dedicated mobile application for vehicle reservation and tracking. In the face of previously manual quotation management and the absence of centralized customer relationship tracking, the objective was to implement a unified digital ecosystem.

The adopted methodology relies on the Unified Process associated with UML (UP/UML) for the rigorous modeling of requirements and business processes. The software solution is built upon a modern 3-tier client-server architecture: a responsive web interface designed with React.js and CSS for all multi-sector activities of the company (vehicle rental, trading, technical services, renewable energies, agropastoral, real estate); a native mobile application developed with React Native and Expo specialized in real-time vehicle booking; and a centralized backend powered by Node.js/Express coupled with a MySQL database managed via the Sequelize ORM.

The results achieved deliver instantaneous synchronization between web and mobile: automated generation of official 2-page pro-forma quotations in PDF format, synchronized multi-item shopping cart, email notifications (Brevo), and a comprehensive administrative dashboard. This solution guarantees SOUTARAH GROUP increased profitability, optimal commercial responsiveness, and complete operational traceability.

GENERAL INTRODUCTION

The service sector is an essential driver of the Ivorian economy, contributing significantly to GDP and employment, with diverse activities ranging from transportation to international trading, technical engineering, and energy. However, these companies face organizational challenges: paper-based management, slowness in quote generation, and lack of customer traceability. In this context, the integration of ICT becomes a strategic imperative for competitiveness. It is within this perspective that the project carried out at SOUTARAH GROUP takes place.

The project title is: « **Design and development of a digital platform for managing the services of SOUTARAH GROUP integrating a mobile application dedicated to vehicle reservation** ».

The project is structured around two complementary components : a web platform and a mobile application. The web platform centralizes the company's six business divisions (vehicle rental, trading/import-export, civil engineering technical services, renewable energies, agro-pastoral, and real estate), as well as catalog management, pro-forma quote generation, and the administration back-office. The mobile application, for its part, meets the specific mobility needs of customers by offering vehicle fleet browsing, availability checking, and direct reservation.

This report provides an account of the entire software engineering process followed and is structured around four major parts:

- Part One presents the project framework and context, introducing the host organization SOUTARAH GROUP, the analysis of the existing system, the targeted objectives, the project specifications, and task scheduling;
- Part Two is dedicated to the conceptual study, detailing the UP/UML methodological approach and the various design diagrams (use cases, activity, sequence, and classes);
- Part Three develops the technical study, justifying the technological choices (React, React Native, Node.js, MySQL) and explaining the overall software architecture;
- Part Four describes the practical implementation of the web platform and mobile application, detailing the developed features, validation testing, achieved results, and financial analysis of the project.



PART I

PROJECT FRAMEWORK AND CONTEXT

CHAPTER 1

OVERVIEW OF THE HOST ORGANIZATION

I. PRESENTATION OF SOUTARAH GROUP

1. General Presentation

SOUTARAH GROUP is a versatile company offering a diversified range of services aimed at both corporate clients and individuals. Thanks to its multi-sector expertise, the company provides its clientele with integrated solutions tailored to diverse needs, while maintaining a uniform standard of quality from a single service provider.

SOUTARAH GROUP's activities span several areas, notably services related to vehicle rental, trading and import-export, as well as project support and execution. This diversity constitutes one of the main strengths of SOUTARAH GROUP. It allows the company to support its clients across various projects and offer them solutions that match their specific requirements.

In an environment marked by the constant evolution of customer needs and the development of digital technologies, SOUTARAH GROUP aims to revolutionize the service delivery industry. The company seeks to strengthen its market position through its multi-sector expertise, constant innovation, and steadfast commitment to customer satisfaction.

2. Mission, Vision and Values

❖ Mission

The mission of SOUTARAH GROUP is to reinvent quality, reliable and tailored services. This mission reflects the company's commitment to offering its clients solutions adapted to their needs, while constantly seeking to improve the quality and reliability of its services.

❖ Vision

The vision of SOUTARAH GROUP is to be the preferred partner of our clients with innovative solutions, exceptional responsiveness and quality services.

❖ Values (S.A.P.E)

The values of SOUTARAH GROUP are grouped around the acronym S.A.P.E :

S — Sustainable and innovative solution

The company prioritizes the implementation of sustainable and innovative solutions to effectively respond to the needs of its clients.

A — Adaptability

SOUTARAH GROUP adapts to the different needs of its clients as well as to changes in its professional environment.

P — Customer priority

Customer satisfaction constitutes an essential element in the design and delivery of the services offered.

E — Staff efficiency

The company values the skills and efficiency of its staff to guarantee the quality of delivered services.

3. Services

SOUTARAH GROUP operates in several business divisions.

- **Vehicle rental** : The vehicle rental service offers mobility solutions adapted to different customer needs, whether for one-time trips or for longer periods.
- **Trading / Import-Export** : Facilitate the supply and distribution of various goods.
- **Technical** : Installation, maintenance and support of equipment and facilities.
- **Renewable energies** : Optimize energy consumption and promote the use of renewable energies.
- **Agropastoral** : Agricultural production and livestock farming.
- **Real estate** : Sale, rental and management of properties.



4. Partnerships

Soutarah Group maintains collaborations with several companies including:

- BESSAC
- DM Company
- CIM IVOIRE
- Southcomp Polaris
- Enabel

CHAPTER 2

PROJECT OVERVIEW

I. PROJECT CONTEXT AND JUSTIFICATION

The continuous growth of SOUTARAH GROUP's activities and the diversification of its divisions generated a substantial volume of quotation requests and reservations. Faced with this increased workload, traditional manual handling mechanisms (isolated emails, informal phone calls, maintaining physical logs) demonstrated their operational limitations. The lack of real-time visibility into rental vehicle availability and the lack of synchronization between sales and technical departments motivated management to initiate the full digital transition of its commercial ecosystem.

II. EXISTING SYSTEM STUDY AND LIMITATIONS

The preliminary analysis of the existing infrastructure revealed that SOUTARAH GROUP had only a static showcase website. This website presented the company but lacked any dynamic database management component, client authentication portal, or automated pricing engine. Inquiries submitted via basic contact forms arrived in a generic mailbox without traceability, making it impossible to generate reliable statistics and causing delays in issuing official quotations.

III. PROJECT OBJECTIVES

1. General Objective

The overall objective is to design and develop a unified software solution comprising a global service management web platform and a dedicated vehicle reservation mobile application, both synchronized in real time via a secure REST API.

2. Specific Objectives

Operationally, the project aims to:

- Develop a modern web portal interactively presenting all 6 service divisions and the product catalog;
- Develop a native mobile application (Android / iOS) dedicated to live fleet browsing, smooth booking with dynamic pricing, and order status tracking;
- Implement a hybrid multi-item shopping cart (vehicles and physical supplies) with an automated tax calculation engine (18% VAT, 2.5% TDT);
- Automate the generation of official 2-page pro-forma quotations in PDF format (with technical specifications and visuals) ready for printing or electronic signature;
- Develop a centralized administrative dashboard for supervising quotes, vehicles, customers, and notifications;
- Integrate a transactional email notification system.

IV. SYSTEM SPECIFICATIONS

The system specifications formalize the functional and non-functional requirements of the interconnected system:

1. Functional requirements on the Customer side (Web & Mobile) :

- Detailed browsing of the vehicle and product catalog by category;
- Multi-criteria filtering of vehicles (make, category, transmission, air conditioning);
- Selection of rental dates with dynamic calculation according to the destination rate (Abidjan / Outside Abidjan) and driver option;
- Shopping / booking cart management and one-click validation;
- Instant download of the official pro-forma quote in PDF adhering to company legal templates;
- Personal quotation management space;

2. Functional requirements on the Administrator side :

- Overall supervision of key performance indicators (estimated revenue, active reservations);
- Comprehensive vehicle fleet management (addition, modification, maintenance status);
- Comprehensive management of trading goods (adding products, updating prices);
- Customer account management and assignment of partner discounts.

3. Technical requirements and constraints :

- Security;
- Availability and performance;
- Portability: Compatibility with modern web browsers (Chrome, Safari, Edge) and mobile operating systems.

V. TASK PLANNING AND GANTT DIAGRAM

The project took place over a two-month period (from August 3 to October 3, 2026), according to the following chronological breakdown:

1. Planning

No.	Task Description	Start Date	Duration	End Date
1	Initial contact, immersion and project understanding	03/08/2026	5 d	07/08/2026
2	Existing system study and specification drafting	08/08/2026	4 d	11/08/2026
3	Conceptual modeling UP/UML	12/08/2026	6 d	17/08/2026
4	Backend API Node.js / Express and database development	18/08/2026	10 d	27/08/2026
5	Web Platform React.js & CSS development	28/08/2026	12 d	08/09/2026
6	Mobile Application React Native / Expo development	09/09/2026	12 d	20/09/2026

7	Integration testing and corrections	21/09/2026	6 d	26/09/2026
---	-------------------------------------	------------	-----	------------

Table 1 : Forecast Planning and Chronogram of Project Tasks

2. Gantt Diagram

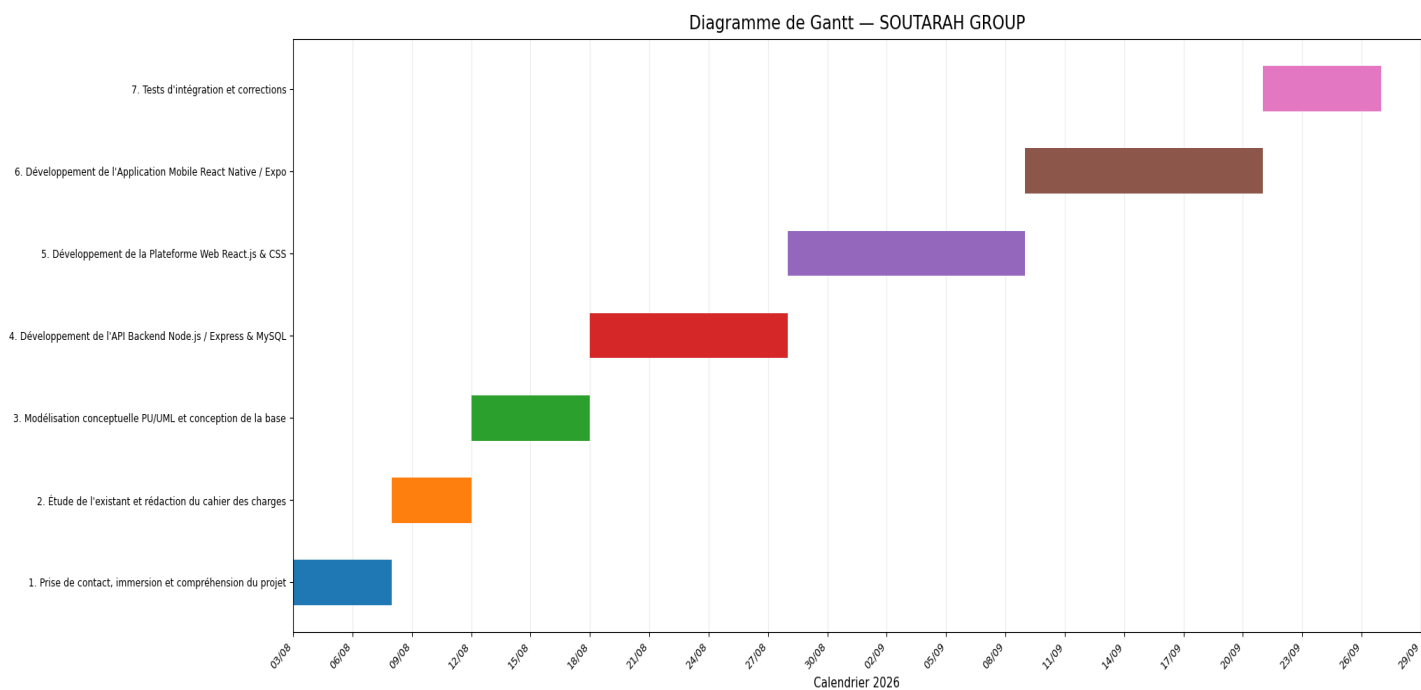


Figure 1 : GANTT Diagram



PART II

CONCEPTUAL STUDY

CHAPTER 3

CHOICE OF ANALYSIS METHODOLOGY

I. COMPARATIVE STUDY OF ANALYSIS METHODOLOGIES

The success of a complex IT project requires a rigorous analysis and modeling phase. Two major methodological approaches were examined: the Cartesian MERISE methodology and the object-oriented UP/UML approach.

1. Overview of the MERISE Methodology

MERISE (Method for Information Systems Design and Development for Enterprises) is an information systems analysis and design methodology. It structures system development by separating data and processing.

The method relies on different levels of abstraction, notably **conceptual, logical, and physical** levels, enabling a progressive transition from requirement specification to system implementation.

For data modeling, MERISE primarily uses the **Conceptual Data Model (CDM)**, which represents entities, their attributes, and relationships. The CDM can then be transformed into a **Logical Data Model (LDM)**, and subsequently into a **Physical Data Model (PDM)** to account for specific DBMS characteristics.

MERISE also models the processes and treatments necessary for system operations. This approach facilitates information organization and overall understanding of the information system prior to its construction.

2. Overview of the UP/UML Methodology

The Unified Process (UP) is an iterative and incremental software engineering process. It organizes project development into multiple phases and enables the system to evolve progressively, from requirements analysis through to deployment.

UP is structured around four primary phases: **inception, elaboration, construction, and transition**. This structure enables better project monitoring, accounts for user feedback, and allows gradual mitigation of risks and errors.

UP is supported by UML (Unified Modeling Language), a standardized visual modeling language used to represent different aspects of a software system. UML facilitates the representation of **actors, functionalities, processes, interactions, and data structures** through various diagrams.

3. Comparative Benchmark : MERISE vs UP/UML

Evaluation Criteria	MERISE Method	Unified Process & UML (UP/UML)
Approach	Sequential, strict data / process separation	Object-oriented, iterative, incremental and user-centered
Data Modeling	CDM, LDM highly adapted to relational SQL databases	Class diagram directly reflecting ORM models
Process Modeling	CPM, LPM focused on information flows	Use case, sequence and activity diagrams
Adaptation to Web / Mobile Architectures	Rigid for event-driven and interactive applications	Perfect fit with modern frameworks (React.js, Node.js, React Native)
System Evolution	Heavy redesign of conceptual models in case of changes	Fluid integration of new features through iterations

Table 2 : Comparative Study between MERISE and UP/UML Methodologies

4. Selection of the Analysis Methodology

For the execution of our project, we selected UML (Unified Modeling Language) associated with the Unified Process (UP). This choice is justified by the specificities of the project and the necessity of clearly representing the diverse capabilities of the system.

- **Alignment with Web and Mobile Architectures :** UML is object-oriented and fits modern software architectures. It facilitates the design of interactions between the frontend, backend, and various system components.
- **Functional Modeling :** UML enables the representation of user interactions with the system, as well as application features through multiple complementary diagrams.
- **System Extensibility :** Thanks to its object-oriented design, UML facilitates **component reusability and scalability**. Models can be incrementally enriched as new features are integrated.

Thus, choosing **UML combined with the Unified Process** provides a methodology tailored to our project, facilitating its analysis, design, and long-term evolution.

II. APPLICATION OF UP/UML TO THE SOUTARAH GROUP SYSTEM

Within the scope of our project, the system incorporates three primary actors :

- **The Visitor / Customer :** browses the catalog, builds a multi-service cart on the Web, makes vehicle reservations on the Mobile application, and downloads PDF quotes;
- **The Administrator :** oversees all operations, manages the fleet, updates pricing, and reviews quotations;
- **The System / REST API :** enforces business rules, verifies availability, and orchestrates notification delivery.

1. Use Case Diagrams

The overall use case diagram illustrates the interactions of actors across both facets of the platform (Web and Mobile).

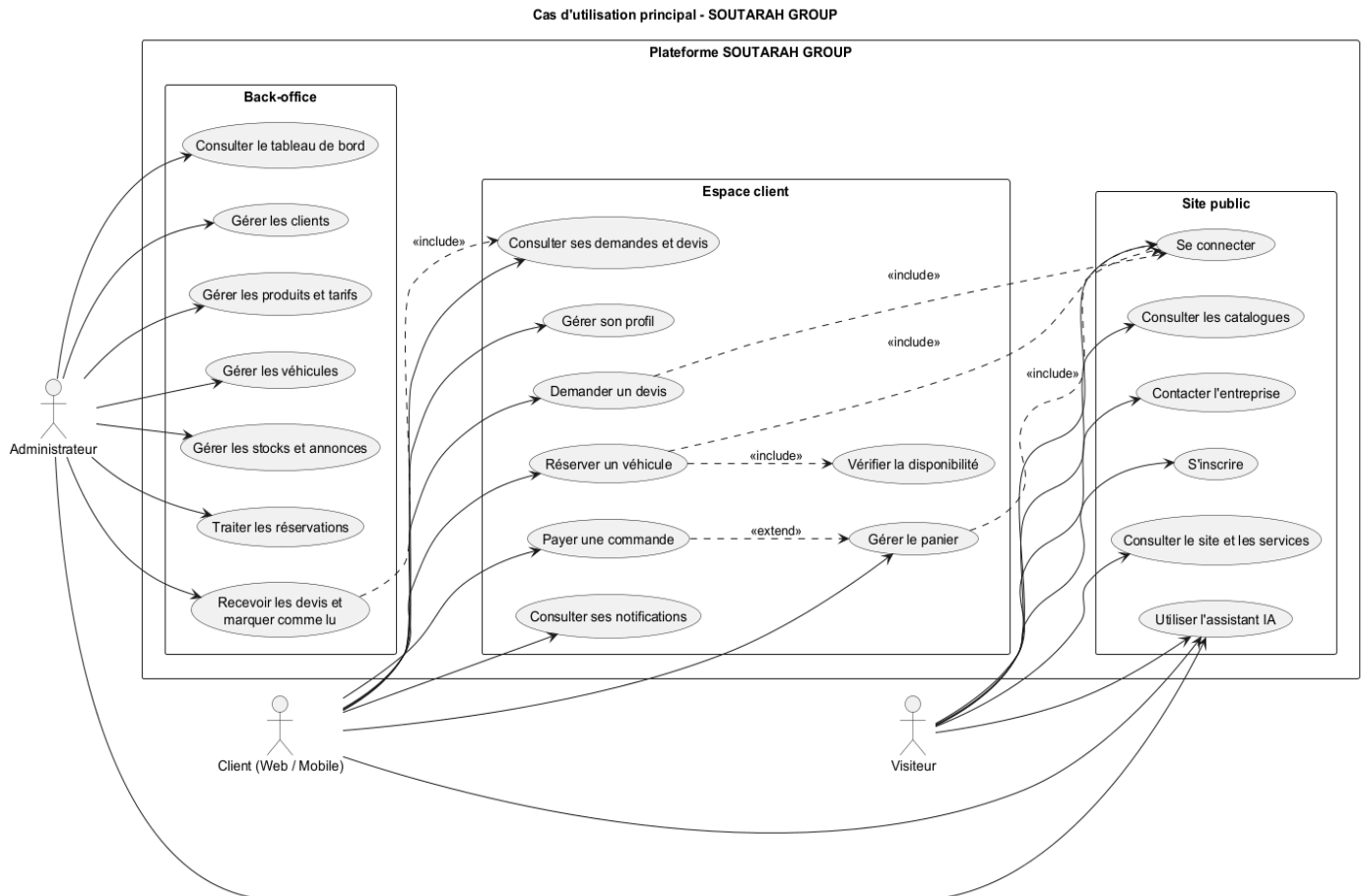


Figure 2 : Global Use Case Diagram (Web & Mobile)

❖ Textual Description of Use Cases

○ Actor : Visitor

Use Case : Browse Site and Services

- **Actors involved :** Visitor, Customer
- **Description :** Allows exploring Soutarah Group activities and catalogs without an account.
- **Preconditions :** Internet connection.
- **Main scenario :**

1. The visitor accesses the website.
2. The visitor browses services, product catalogs, and vehicles.

- **Postconditions** : Information is displayed.

Use Case : Use the AI Assistant

- **Actors involved** : Visitor, Customer
- **Description** : Allows asking questions and being guided automatically.
- **Preconditions** : None.
- **Main scenario** :
 1. The user opens the chat widget.
 2. The user enters a question.
 3. The AI responds instantly.
- **Postconditions** : The user receives the requested information.

Use Case : Register and Log In

- **Actors involved** : Visitor, Customer
- **Description** : Allows creating an account and accessing the personal customer portal.
- **Preconditions** : Have a valid email address.
- **Main scenario** :
 1. The user enters login credentials.
 2. The system verifies and grants access.
- **Postconditions** : The user is authenticated as a Customer.

- **Actor** : Customer (Web / Mobile)

Use Case : Reserve a Vehicle

- **Actors involved** : Customer (Mobile / Web)
- **Description** : Allows selecting and reserving a vehicle from the fleet of 21 vehicles.
- **Preconditions** : The customer must be logged in.

- **Main scenario :**

1. The customer selects a vehicle and rental dates.
2. The system checks availability in real time.
3. The customer confirms the reservation.

- **Postconditions :** The reservation is recorded pending validation.

Use Case : Request a Quotation

- **Actors involved : Customer**

- **Description :** Allows submitting a detailed price estimate request for a service or product.

- **Preconditions :** The customer must be logged in.

- **Main scenario :**

1. The customer selects required items and validates the request.
2. The system generates the downloadable pro-forma quotation.

- **Postconditions :** The quote request is transmitted to management.

Use Case : Pay for an Order

- **Actors involved : Customer**

- **Description :** Allows settling a deposit or paying for an order online.

- **Preconditions :** Have a confirmed reservation or validated cart.

- **Main scenario :**

1. The customer chooses the payment method (Mobile Money or Card).
2. The customer completes the transaction.
3. The system confirms payment receipt.

- **Postconditions :** The payment is validated and a receipt is issued.

- **Actor : Administrator**

Use Case : View the Dashboard

- **Actors involved : Administrator**

- **Description :** Allows monitoring key indicators (turnover, reservations, activities).

- **Preconditions :** The administrator must be authenticated.
- **Main scenario :**
 1. The administrator accesses the back-office.
 2. The system displays statistics and KPIs.
- **Postconditions :** The administrator views the overall state of the platform.

Use Case : Manage Vehicles

- **Actors involved :** Administrator
- **Description :** Allows adding, updating, and tracking the status of the 21-vehicle fleet.
- **Preconditions :** The administrator must be authenticated.
- **Main scenario :**
 1. The administrator navigates to the vehicle fleet list.
 2. The administrator updates availability, rates, or mechanical status.
- **Postconditions :** Fleet details are refreshed on web and mobile platforms.

Use Case : Receive Quotations and Mark as Read

- **Actors involved :** Administrator
- **Description :** Allows viewing received quotation requests and confirming their processing.
- **Preconditions :** The administrator must be authenticated.
- **Main scenario :**
 1. The administrator opens the quotation request list.
 2. The administrator inspects quote contents.
 3. The administrator clicks "Mark as read".
- **Postconditions :** The quote status changes to "Read / In Progress".

2. Activity Diagrams

2.1. Activity Diagram : Customer Account Creation and Management

The following activity diagram details the vehicle reservation process flow, highlighting algorithmic anti-collision date controls.

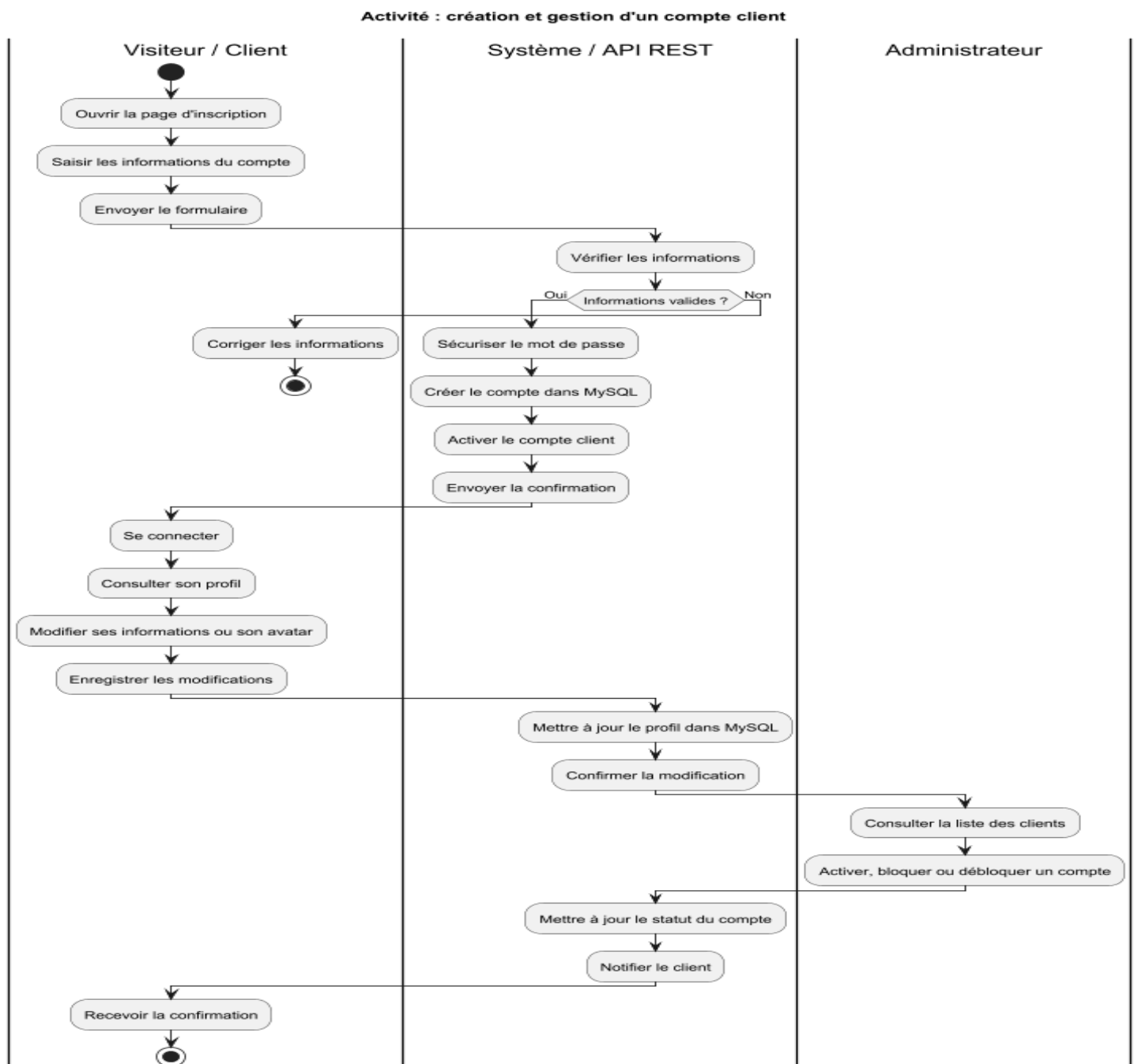


Figure 3 : Activity Diagram : Customer Account Creation and Management

2.2. Activity Diagram : Vehicle Reservation

The Vehicle Reservation activity diagram illustrates the process through which a customer rents a vehicle on the SOUTARAH platform. The process begins when the customer selects a vehicle and specifies rental dates. The system then verifies availability: if unavailable, the customer is notified and can modify dates or choose another vehicle. If available, the vehicle is added to the cart. When the customer checks out the cart, the system automatically generates the quotation and redirects to the order page. The customer then selects a payment method (online via Card/Mobile Money or cash upon delivery) and confirms the order. The system automatically validates the quotation, records the reservation in the database, and transmits the order to the administrator. The latter does not need to approve the quote manually: they receive the notification, review order details, and simply mark the quote as read to prepare the vehicle. Finally, the customer receives booking confirmation. This diagram thus highlights the nominal order flow and simplified processing on the administrative side.

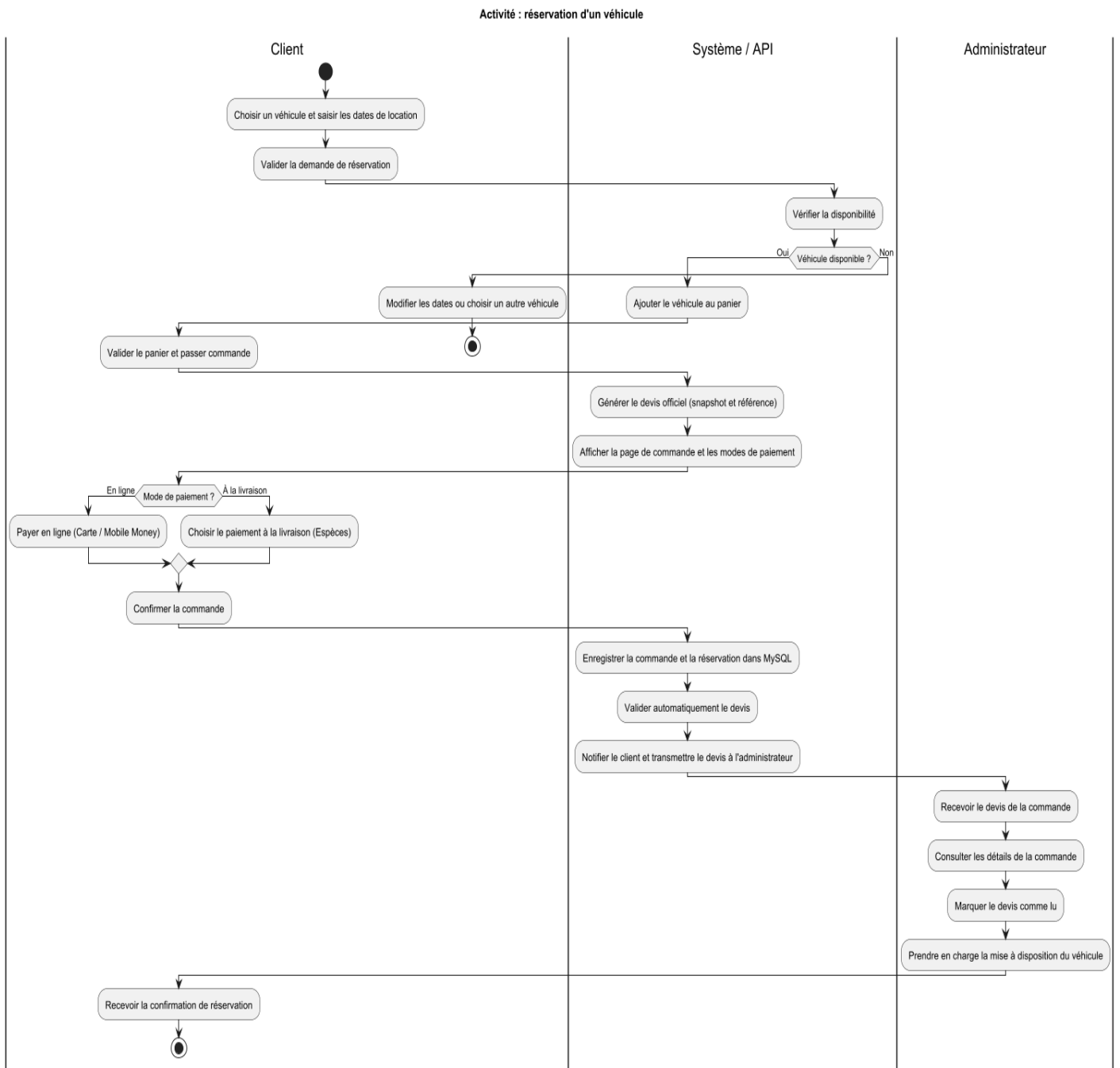


Figure 4 : Activity Diagram : Vehicle Reservation

2.3. Activity Diagram : Product Purchase

The Product Purchase activity diagram illustrates the process through which a customer purchases an item on the SOUTARAH platform. The process begins when the customer browses the catalog, selects a product, and adds it to their cart. The system immediately calculates and displays in the cart the detailed amounts (excluding taxes, VAT, and total including taxes). When the customer checks out the cart and proceeds to order placement, the system generates the official time-stamped quotation and displays available payment methods. The customer chooses their payment method, either online (via credit card or Mobile Money) or in cash upon delivery, then confirms the order. The system subsequently records the order in the database, automatically validates the quotation, and alerts the administrator. The administrator inspects order details and simply marks the quote as read to initiate item preparation and shipping. Finally, the customer receives order confirmation. This diagram thus underscores the direct purchasing flow and transparency of calculations starting right from the shopping cart stage.

Activité : achat d'un produit

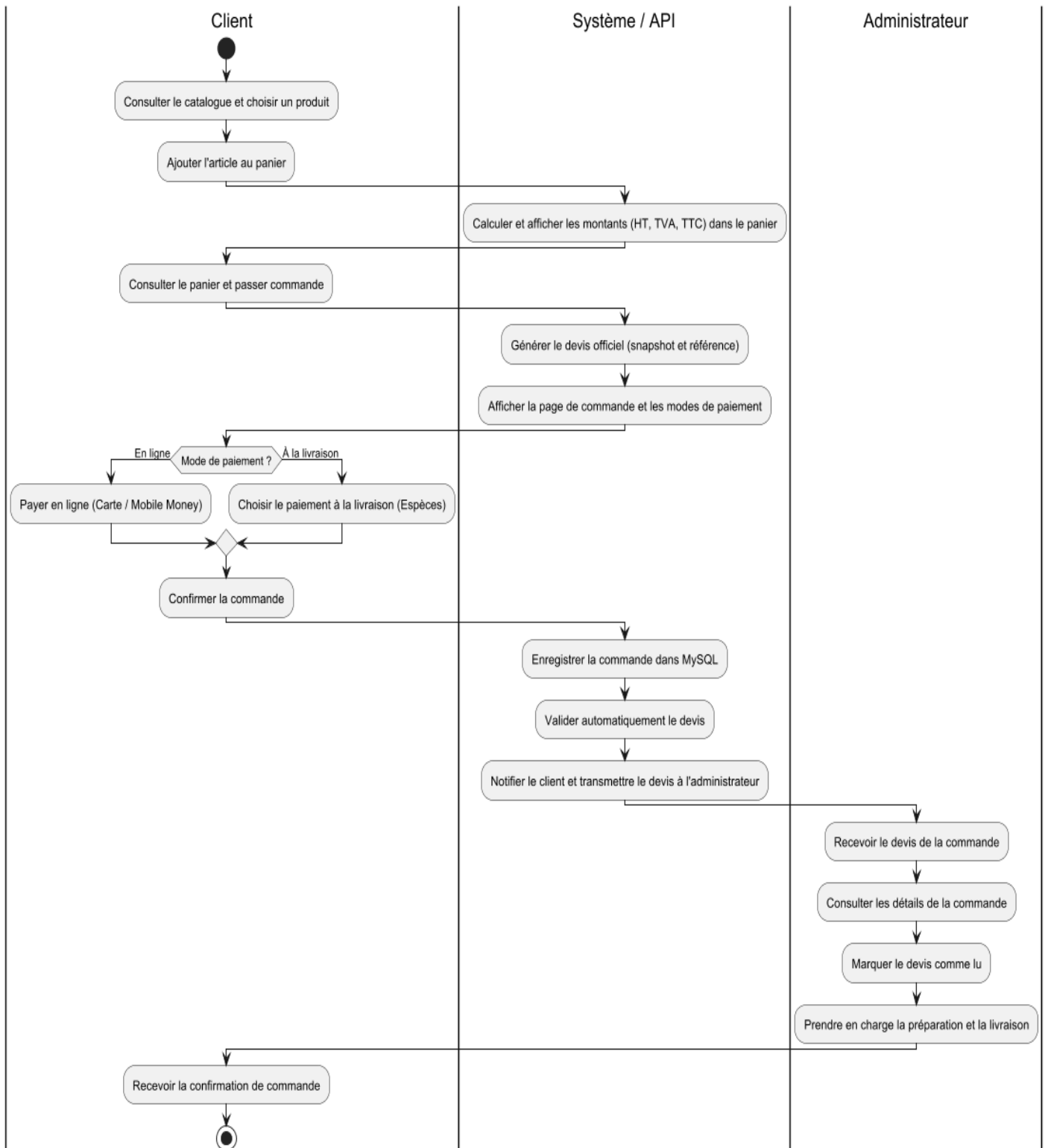


Figure 5 : Activity Diagram : Product Purchase

3. Sequence Diagrams

3.1. Sequence Diagram : Online Payment

This sequence diagram illustrates the flow of an electronic financial transaction carried out via the Genius Pay gateway (integrating bank cards and local Mobile Money solutions). Upon cart confirmation, the API server initializes a secure checkout session and redirects the client to the external payment portal. Following authorization and bank debit, the payment gateway notifies the API server via a secure webhook to update the order status to paid in the database. The Brevo service immediately dispatches an electronic receipt to the customer, while the web interface presents the payment confirmation screen.

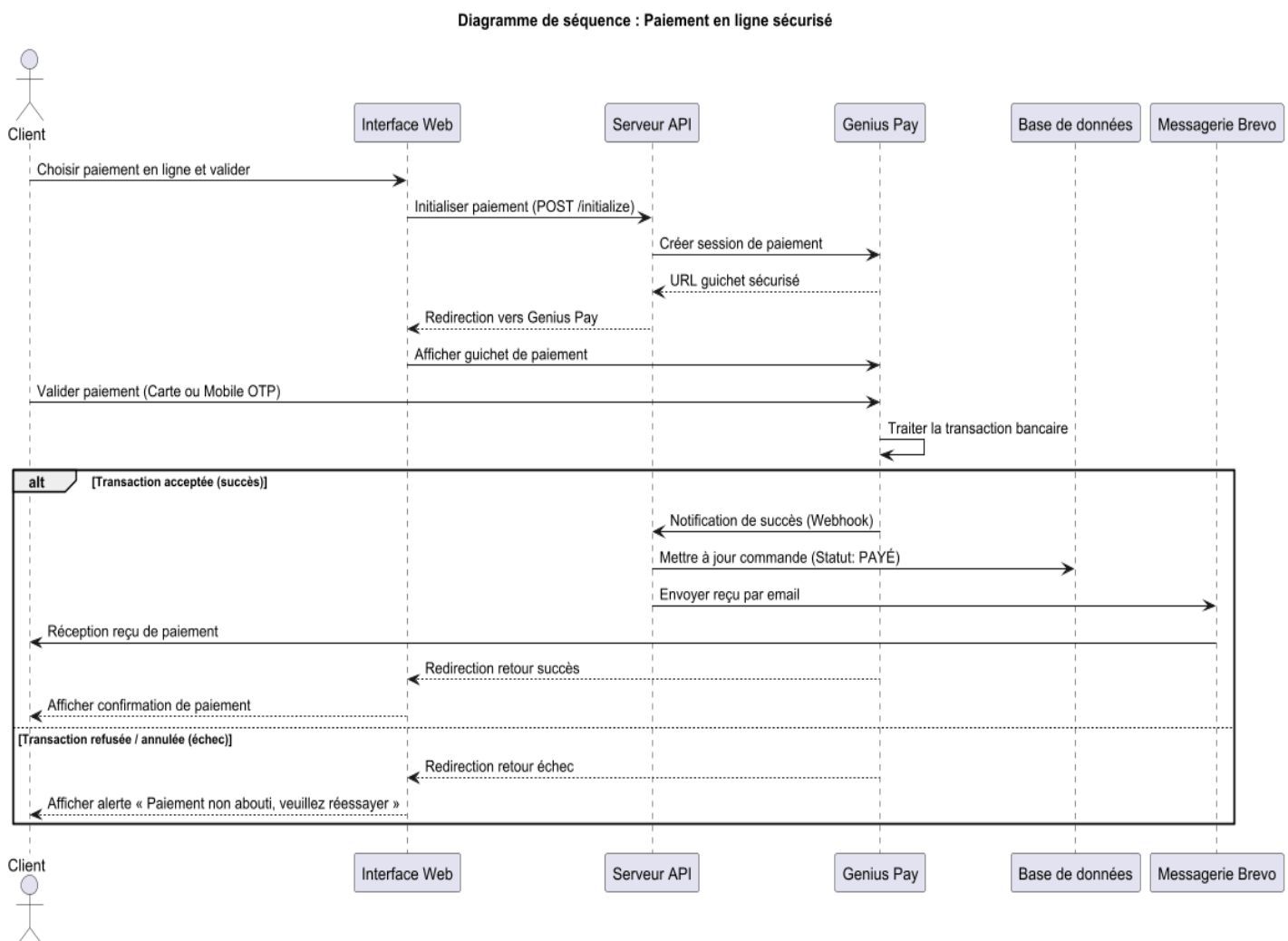


Figure 6 : Sequence Diagram : Online Payment

3.2. Sequence Diagram : Vehicle Reservation

This sequence diagram illustrates the complete workflow of reserving a vehicle on the platform. When the customer selects a car and a rental period, the web interface queries the API server to verify the absence of scheduling conflicts in the database. Once availability is confirmed and the order is validated by the client, the API records the reservation along with the associated quotation, then triggers the Brevo messaging service to instantly dispatch confirmation emails to both customer and administrator. In their management dashboard, the administrator accesses the order details sheet and marks the quote as read, initiating operational vehicle preparation without requiring manual quote validation.

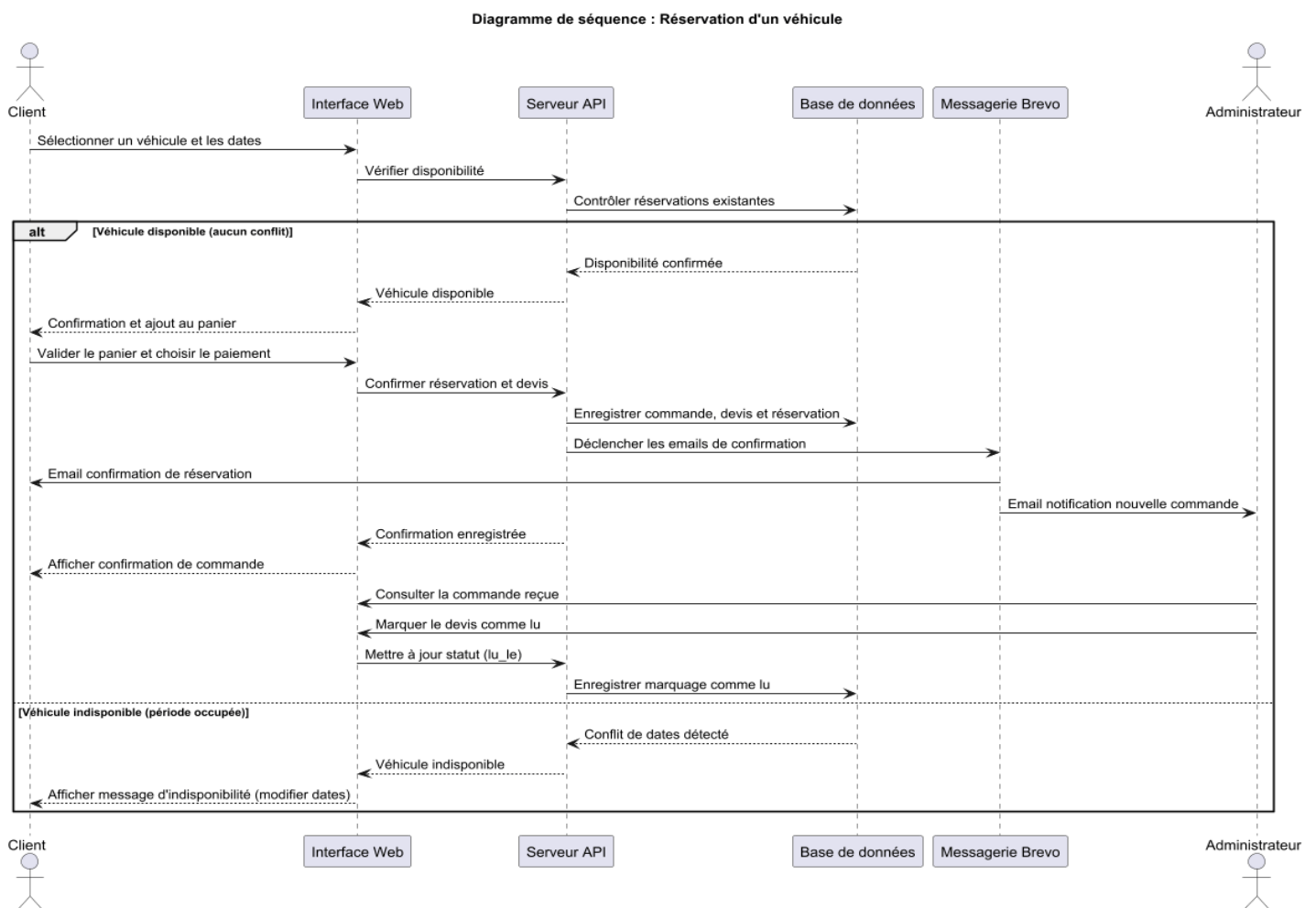


Figure 7 : Sequence Diagram : Vehicle Reservation

3.3. Sequence Diagram : Product Purchase

The Product Purchase sequence diagram describes the acquisition workflow for goods from the trading catalog. Upon order validation, data is transmitted to the API server, which logs the transaction into the database and automatically validates the corresponding quote. The Brevo service is then invoked to dispatch the purchase summary to the customer and alert the administrator of the new order. The administrator reviews ordered item specifications on their dashboard and marks the quote as read to schedule package preparation and delivery.

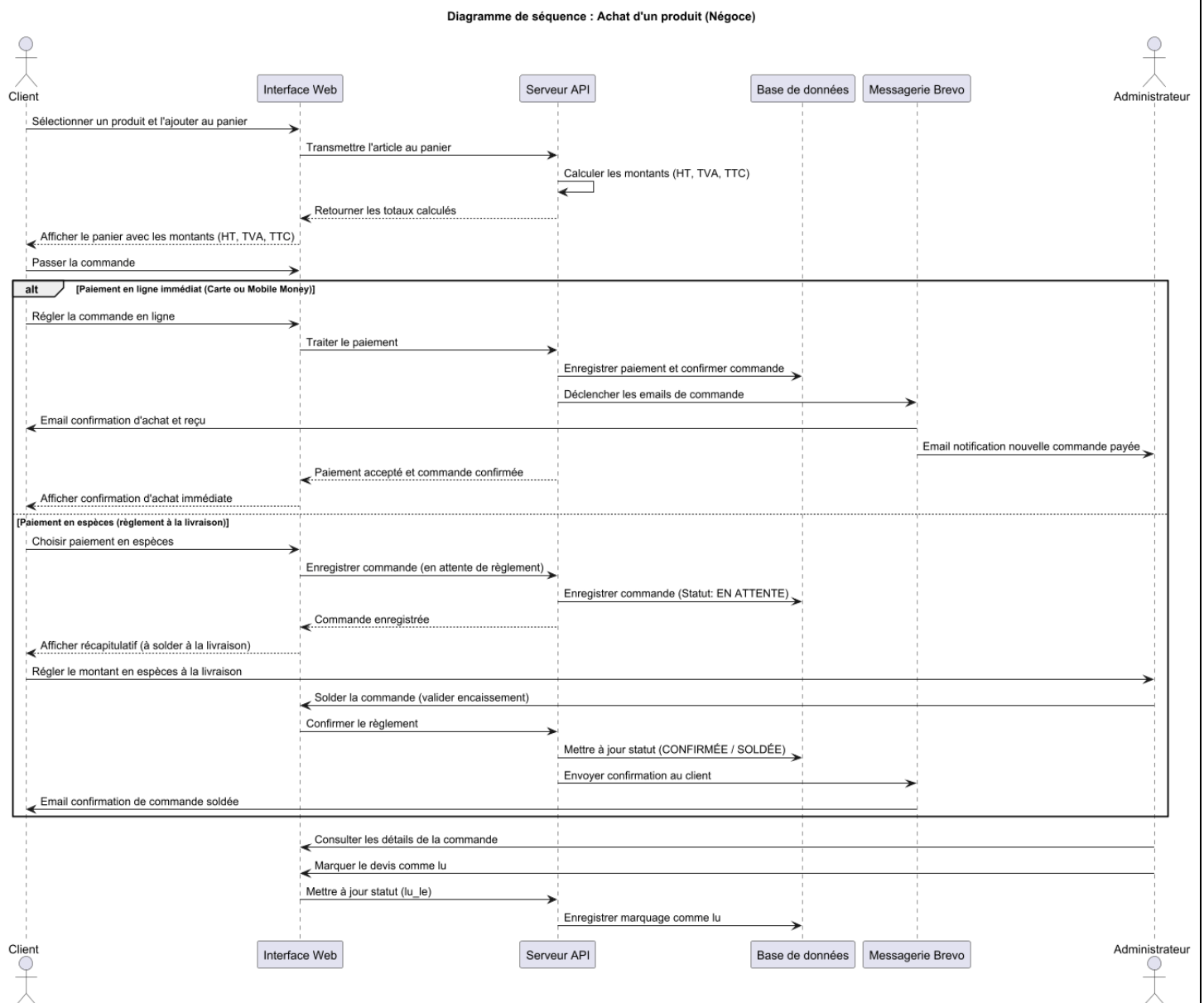


Figure 8 : Sequence Diagram : Product Purchase

3.4. Sequence Diagram : User Registration

This sequence diagram describes the registration process for a new user on the SOUTARAH platform. The visitor inputs their information (individual or corporate entity) through the web interface form. This data is transmitted to the API server, which first verifies the uniqueness of the email address and phone number against the database. After confirming the absence of duplicates, the API server secures the password via Bcrypt hashing and inserts the user and client profile records into the database. The API server then invokes the Brevo service to dispatch a welcome and confirmation email to the visitor, while the web interface confirms account creation and redirects them to the login page.

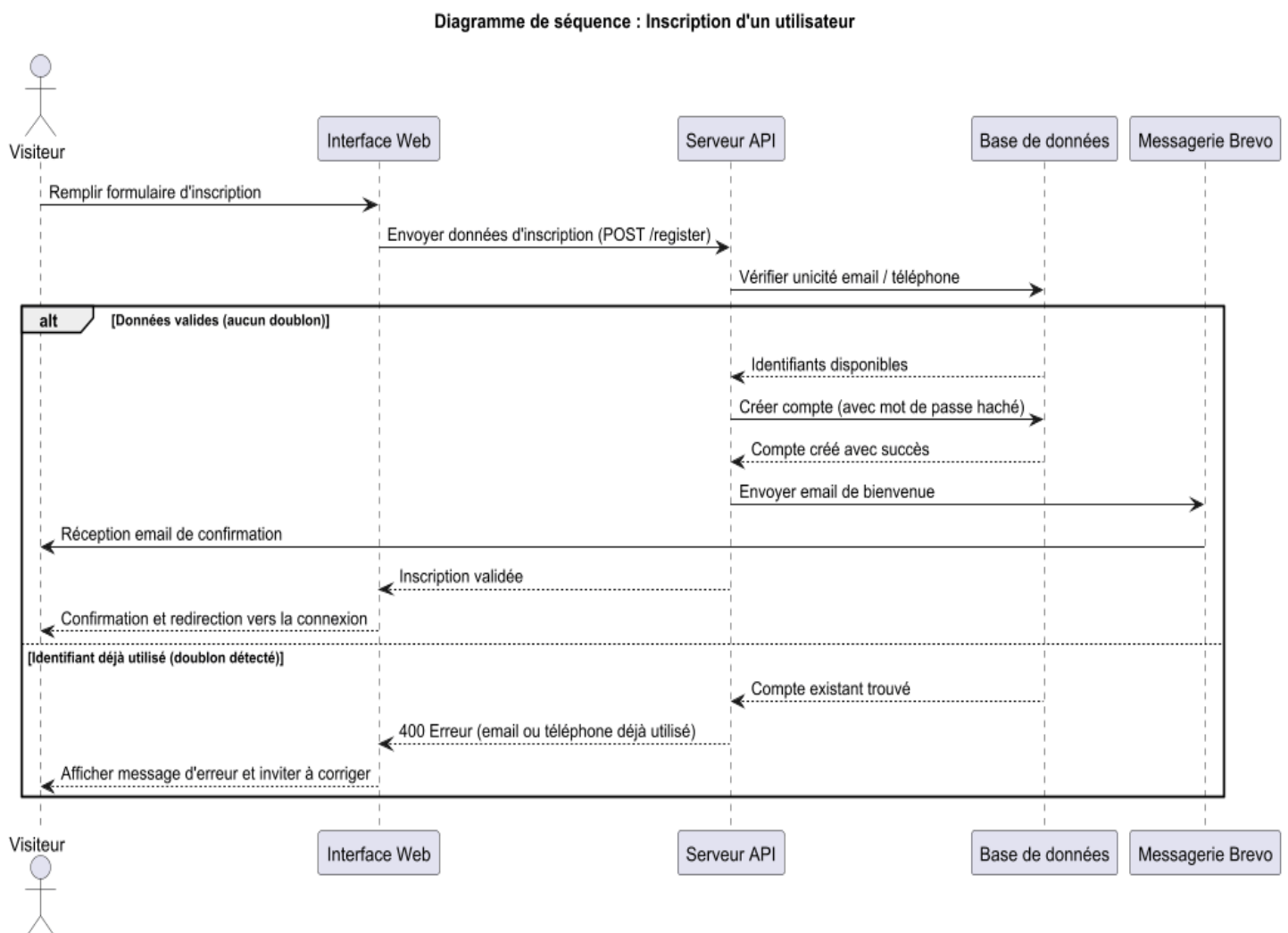


Figure 9 : Sequence Diagram : User Registration

4. Class Diagram

It represents the static structure of the system by describing classes, their attributes, methods, and relationships connecting them. It is generally constructed following an in-depth understanding of system requirements and interactions, serving as the blueprint for the implementation phase.

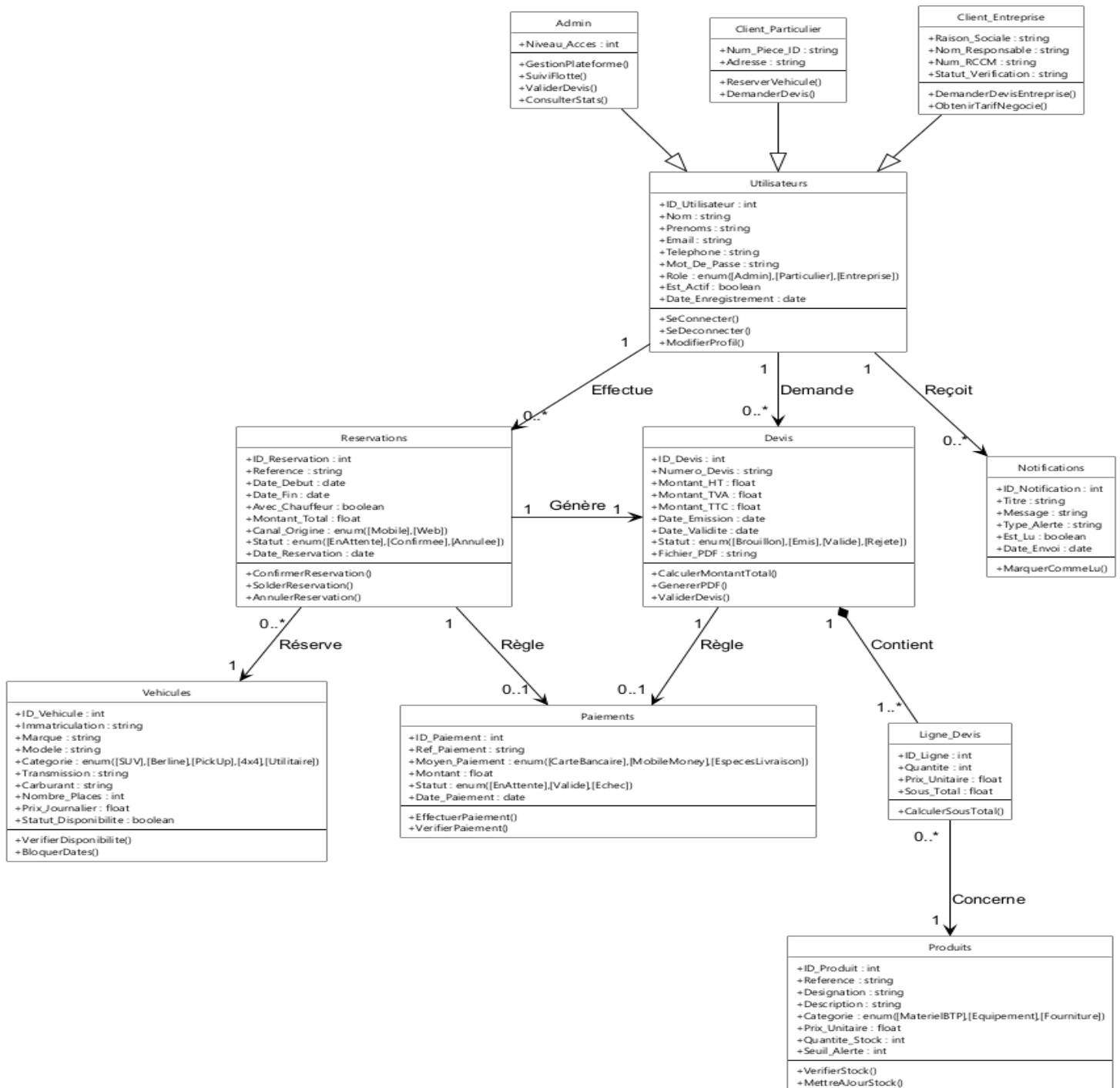


Figure 10 : System Class Diagram



PART III

TECHNICAL STUDY

CHAPTER 4

SYSTEM DESIGN AND TECHNOLOGICAL CHOICES

I. ANALYSIS AND JUSTIFICATION OF TECHNOLOGICAL CHOICES

The choice of technologies and tools represents a decisive phase in project implementation. To guarantee the relevance and effectiveness of decisions made, an in-depth analysis of solutions available on the market was conducted, taking into account project specifications, our skills, and long-term objectives. This process was grounded in several essential selection criteria: performance and scalability, cost and licensing conditions, security, integration flexibility, as well as documentation quality and technical support accessibility. These factors guided technological choices toward robust, tailored, and sustainable solutions.

1. Integrated Development Environment (IDE)



Visual Studio Code was selected for its lightweight footprint, native support for the JavaScript/TypeScript ecosystem, its extensions for React, React Native, and MySQL, as well as its integrated terminal.

Figure 11 : Visual Studio
Code Editor Logo

2. UI/UX Mockups

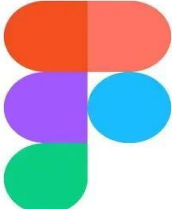


Figure 12 : Figma
Logo

Figma is an online UI/UX interface design tool used to create interactive and collaborative mockups. Accessible from a simple browser, it allows teams to design and share visually clear prototypes. In a web project, it facilitates interface visualization prior to development.

3. Frontend Web Technologies

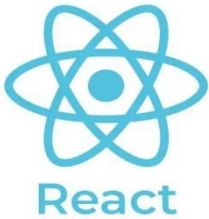


Figure 13 : React
Framework Logo

For the web platform, the React.js framework coupled with CSS was chosen. Its architecture based on the Virtual DOM and reusable components provides a reactive and modular interface, facilitating seamless navigation across the catalog, interactive cart, and administrator dashboard.

4. Mobile Technologies

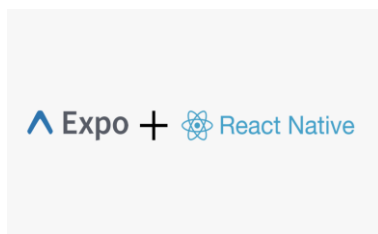


Figure 14 : React Native & Expo
mobile framework logo

For the mobile application, React Native combined with the Expo ecosystem was retained. This technology offers a single codebase in TypeScript/JavaScript natively compiled for Android and iOS, guaranteeing high performance, convenient access to device hardware features (file system, PDF sharing), and a substantial reduction in development costs.

5. Backend Technologies and API



Figure 15 : Node.js
Logo

The application server relies on Node.js and the Express.js framework. This choice enables maintaining the same language (JavaScript/TypeScript) across the entire software pipeline. The stateless REST architecture based on JSON exchanges and secured by JWT tokens ensures optimal scalability and uniformly powers both the Web and Mobile applications.

6. Notification and Messaging Technologies



Figure 16 : Brevo
Logo

To manage notifications and electronic messaging delivery, Brevo was selected. It allows automating email dispatches linked to various platform events, notably reservation confirmations, order alerts, and messages intended for customers.

Its API integration centralizes and automates communication with users, while facilitating interoperability across web and mobile applications.

6. Online Payment Technologies



Figure 17 : Genius Pay
Logo

To secure financial transactions, Genius Pay was selected. It allows automating the settlement of reservations and orders by integrating bank cards and Mobile Money.

Its integration via API and Webhook ensures instant, reliable and secure payments across web and mobile applications.

7. Deployment Technologies



Figure 18 : Hostinger
Logo

To host the platform online, Hostinger was selected.

The selection of Hostinger is explained particularly by its ease of deployment, accessible cost, and compatibility with the technologies used in the project.

II. DATABASE MANAGEMENT SYSTEM (DBMS)

The selection of the Database Management System (DBMS) represents an important step in application design and deployment. It must fulfill system requirements regarding performance, security, reliability, scalability, and ease of maintenance. In the context of the SOUTARAH GROUP project, several solutions were studied, notably **MySQL**, **PostgreSQL**, **SQL Server**, **Oracle**, and **MongoDB**. The analysis evaluated their data models, performance, costs, and integration capabilities.

1. Comparison of Major DBMS

Criteria	MySQL	PostgreSQL	SQL Server	Oracle	MongoDB
Type	Relational, open source	Relational, open source	Relational, proprietary	Relational, proprietary	NoSQL, document-oriented
Data model	Relational, structured	Relational, structured	Relational, structured	Relational, structured	Flexible documents
Performance	Very good	Very good	Very good	Excellent	Excellent on certain large volumes
Scalability	Good	Very good	Very good	Excellent	Excellent

Cost	Free	Free	Paid	Paid	Free / cloud offerings
Ease of use	High	Medium to high	High	More complex	High
Compatibility with selected hosting	Yes	Not available on selected plan	No	No	No

Table 3 : Comparative Analysis of Database Management Systems

2. Selection of the DBMS

After comparing the various solutions, MySQL was chosen as the database management system for our project.

This choice is primarily justified by the nature of data handled by the platform. The application must manage structured records such as **customers, vehicles, reservations, products, orders, and quotations.**

Another determining factor in this selection involves the production environment. The hosting package acquired by SOUTARAH GROUP specifically supports **MySQL**. Choosing this DBMS ensures optimal compatibility between the developed application and the hosting infrastructure.

Finally, MySQL benefits from a widely adopted ecosystem in web development and integrates seamlessly with the technologies used in the project. It thus represents a **reliable, high-performance, accessible, and suitable solution.**

III. GENERAL SYSTEM ARCHITECTURE

Our solution is based on a distributed three-tier (3-tier) architecture:

- 1. Presentation Tier (Clients) :** Users interact with the system via two complementary channels :

- A **mobile application (React Native / Expo)** dedicated specifically to customers for browsing the vehicle fleet and making reservations online;
 - A **web platform (React / Vite)** enabling overall service management, quotation requests, and administrative oversight (Back-Office).
2. **Network Tier (Internet)** : Ensures secure communication between clients and the server via HTTPS and REST API requests exchanging JSON data.
3. **Business and Data Tier (Server & DB)** : A **Node.js / Express** application server manages business logic (automated availability verification preventing double bookings, quotation calculations, JWT security) and communicates with a **MySQL** database for data persistence.

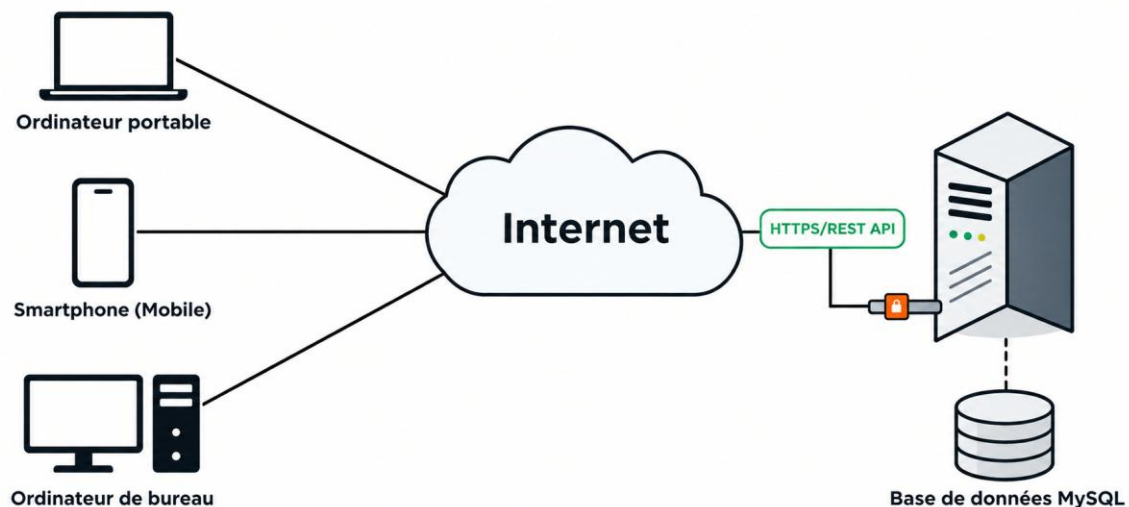


Figure 19 : General System Architecture Diagram



PART IV

IMPLEMENTATION

CHAPTER 5

DEVELOPMENT OF THE PLATFORM AND MOBILE APPLICATION

I. MYSQL DATABASE SCHEMA IMPLEMENTATION

The MySQL database was designed to guarantee transactional consistency across data flows. It incorporates the following main tables:

- Users & Clients: credential management (email, hashed password, 'CLIENT' or 'ADMIN' role) and links to individual or corporate profiles;
- Vehicles: make, model, category, daily rental rate, status, photographs;
- Reservations: start and end dates, with/without driver option, total amount, status;
- Services & Products: general catalog of the 6 divisions with technical specifications and pricing;
- Quotations: unique reference (e.g.: DMD-2026-XXXX), creation timestamp, amount excl./incl. tax;
- Notifications: alerts dispatched to clients and administrators.

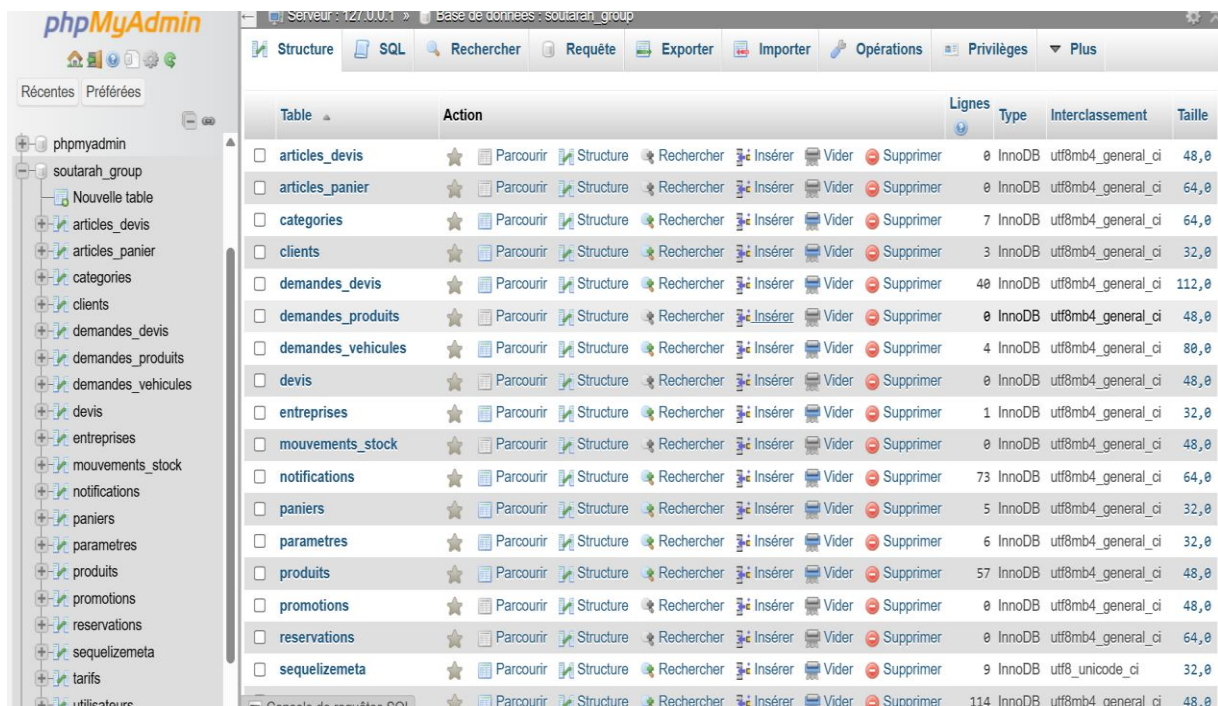


Table	Action	Lignes	Type	Interclassement	Taille
articles_devis	Parcourir Structure Rechercher Insérer Vider Supprimer	0	InnoDB	utf8mb4_general_ci	48,0
articles_panier	Parcourir Structure Rechercher Insérer Vider Supprimer	0	InnoDB	utf8mb4_general_ci	64,0
categories	Parcourir Structure Rechercher Insérer Vider Supprimer	7	InnoDB	utf8mb4_general_ci	64,0
clients	Parcourir Structure Rechercher Insérer Vider Supprimer	3	InnoDB	utf8mb4_general_ci	32,0
demandes_devis	Parcourir Structure Rechercher Insérer Vider Supprimer	40	InnoDB	utf8mb4_general_ci	112,0
demandes_produits	Parcourir Structure Rechercher Insérer Vider Supprimer	0	InnoDB	utf8mb4_general_ci	48,0
demandes_vehicules	Parcourir Structure Rechercher Insérer Vider Supprimer	4	InnoDB	utf8mb4_general_ci	80,0
devis	Parcourir Structure Rechercher Insérer Vider Supprimer	0	InnoDB	utf8mb4_general_ci	48,0
entreprises	Parcourir Structure Rechercher Insérer Vider Supprimer	1	InnoDB	utf8mb4_general_ci	32,0
mouvements_stock	Parcourir Structure Rechercher Insérer Vider Supprimer	0	InnoDB	utf8mb4_general_ci	48,0
notifications	Parcourir Structure Rechercher Insérer Vider Supprimer	73	InnoDB	utf8mb4_general_ci	64,0
paniers	Parcourir Structure Rechercher Insérer Vider Supprimer	5	InnoDB	utf8mb4_general_ci	32,0
parametres	Parcourir Structure Rechercher Insérer Vider Supprimer	6	InnoDB	utf8mb4_general_ci	32,0
produits	Parcourir Structure Rechercher Insérer Vider Supprimer	57	InnoDB	utf8mb4_general_ci	48,0
promotions	Parcourir Structure Rechercher Insérer Vider Supprimer	0	InnoDB	utf8mb4_general_ci	48,0
reservations	Parcourir Structure Rechercher Insérer Vider Supprimer	0	InnoDB	utf8mb4_general_ci	64,0
sequelizemeta	Parcourir Structure Rechercher Insérer Vider Supprimer	9	InnoDB	utf8mb4_general_ci	32,0
utilisateurs	Parcourir Structure Rechercher Insérer Vider Supprimer	114	InnoDB	utf8mb4_general_ci	48,0

Figure 20 : Structure and Relational Schema of MySQL Database

II. IMPLEMENTATION OF THE AI ASSISTANT AND TRANSACTIONAL EMAIL NOTIFICATION SERVICE (BREVO / SMTP)

To modernize customer engagement and streamline operational communications at SOUTARAH GROUP, we enriched the platform with two modules : an intelligent conversational agent and an automated transactional email service.

1. Intelligent Virtual Assistant (AI Agent)

To guide visitors and assist managers, a virtual agent based on an advanced Large Language Model (LLM) was developed:

- **Technical Architecture :** It is built around a floating interactive widget in React connected via a secure REST API to the backend service. The latter queries the OpenAI model (gpt-4o-mini) by dynamically injecting real-time context from the MySQL database (available vehicles, technical specifications, prices, and stock levels).
- **Customer Role :** The assistant understands inquiries expressed in natural language (e.g.: "I am looking for a 5-seater air-conditioned vehicle for Yamoussoukro"), suggests the appropriate model, and generates clickable action buttons to immediately redirect the user to the vehicle sheet or to the shopping cart.
- **Administrator Role :** When authenticated with the ADMIN role, the assistant transforms into a decision-support tool: it executes aggregation queries to answer operational management questions (products near stockout threshold, most rented vehicles, pending quotation volume).

2. Transactional Email and Messaging Service (Brevo / SMTP)

To guarantee traceability and respond promptly to business inquiries, an email notification system was implemented:

- **Delivery Infrastructure :** To send emails, our Node.js server utilizes Nodemailer, which formats messages. These messages are then transmitted to the professional Brevo delivery service, which handles dispatch to recipients. Communication with Brevo is fully encrypted and secure (TLS protocol).
- **Customer Notifications :** The system automatically triggers an email dispatch upon new user registration, upon booking confirmation, and delivers the official 2-page proforma quotation in PDF format as soon as the shopping cart is checked out.

- **Administrator Alerts :** Whenever a customer adds a vehicle reservation or an item to their cart, an immediate alert is transmitted to the sales team containing client contact details, date breakdowns, and estimated totals, enabling proactive customer follow-up.

3. Secure Online Payment Module (Genius Pay)

To secure payments and automate order validation, a payment gateway was integrated via Genius Pay:

- **Infrastructure and Gateway :** Our Node.js server initializes the transaction via the Genius Pay REST API and redirects the customer to its secure, TLS-encrypted check-out portal. No sensitive banking information transits through our servers, ensuring optimal security.
- **Payment Methods :** The platform accepts credit cards (Visa, Mastercard) as well as local Mobile Money. Transactions are protected by strong authentication (OTP/SMS code) prior to effective settlement.
- **Validation and Notifications :** Upon banking validation, Genius Pay notifies our API directly via Webhook to update order status to "PAID" in the database. The Brevo service immediately routes the electronic receipt to the client, while in case of refusal, an alert invites them to retry.

III. IMPLEMENTATION OF WEB PLATFORM INTERFACES

The web portal was developed with a modern aesthetic adhering to SOUTARAH's graphic guidelines:

1. Home and Authentication Pages

- ❖ The homepage serves as the primary storefront. It presents the company history, S.A.P.E values, service divisions, and integrates an intelligent virtual assistant guiding visitors.

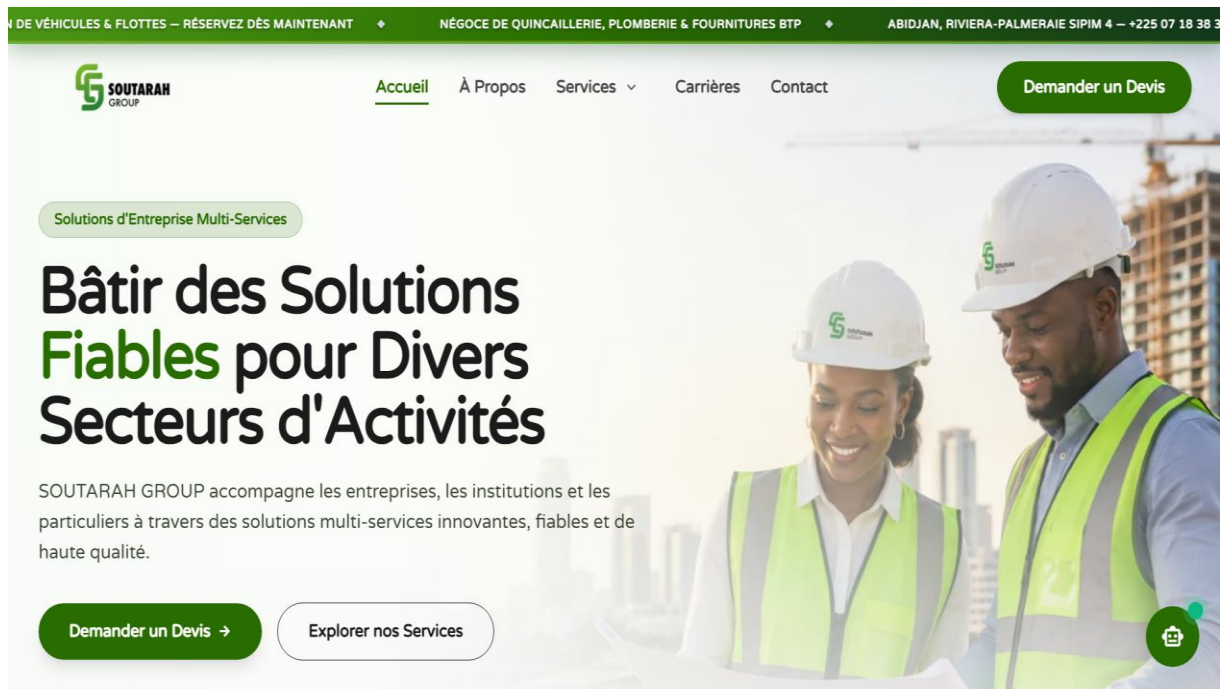


Figure 21 : Web Interface : Home Page

- ❖ The login page allows clients to access their personal portal by entering their login credentials, specifically their email address and password. After credential verification, the client is authenticated and can access features reserved for them.

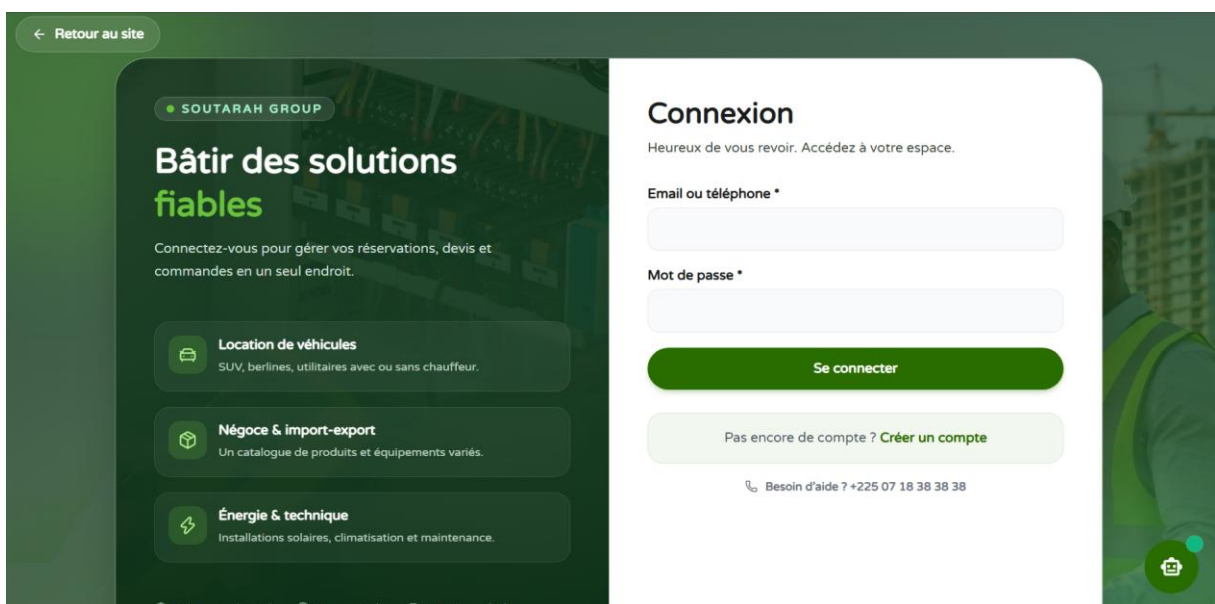


Figure 22 : Web Interface : Login Page and Authentication Form

2. Product Catalog

Allows clients to browse supplies, construction materials, and tools, with direct add-to-cart capabilities.

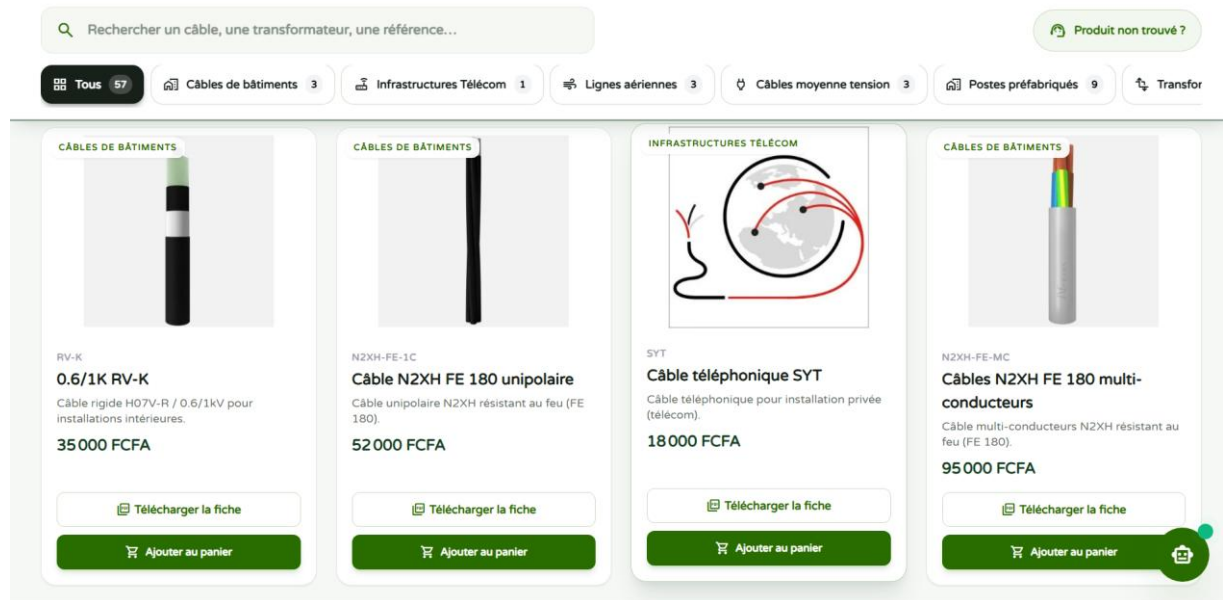


Figure 23 : Web Interface : Product Catalog

3. Vehicle Reservation Module

Enables vehicle selection, precise choice of pick-up and return dates, and the option with driver.

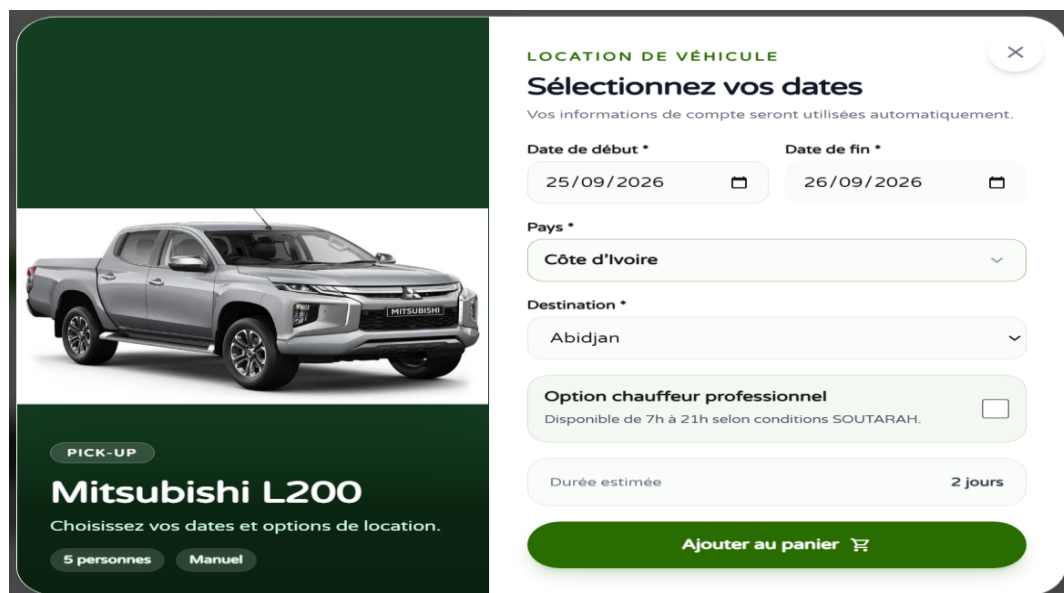


Figure 24 : Web Interface : Vehicle Reservation Module

4. Multi-Item Cart and PDF Quotation Generation

The shopping cart combines both vehicle rentals and physical goods, calculates statutory taxes, and generates the instantly downloadable pro-forma quote.

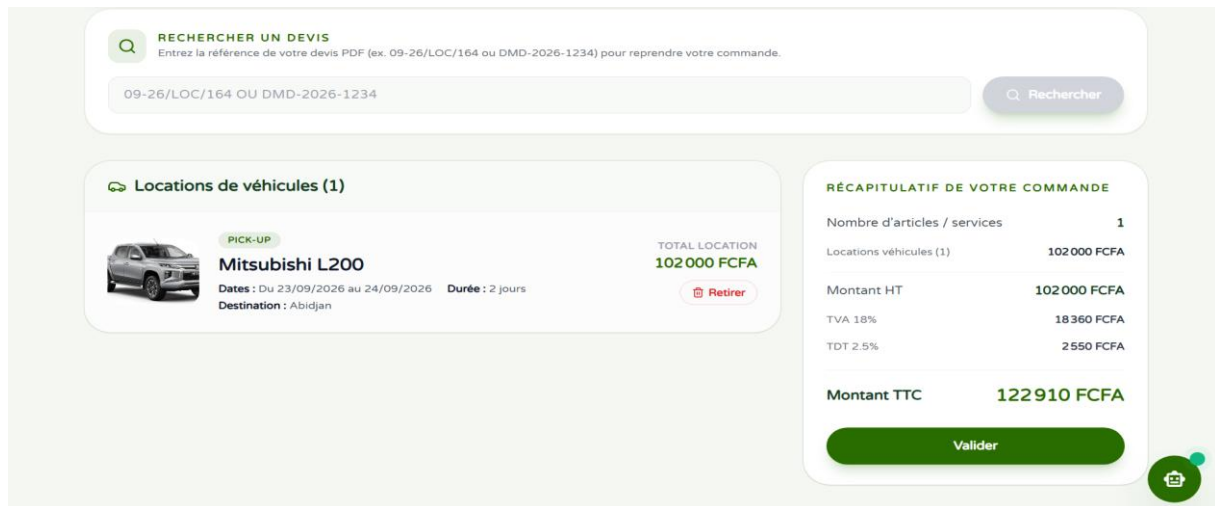


Figure 25 : Web Interface : Multi-Item Cart

5. Customer Portal and Request Tracking

Provides authenticated users with an overview of their pending quotations.

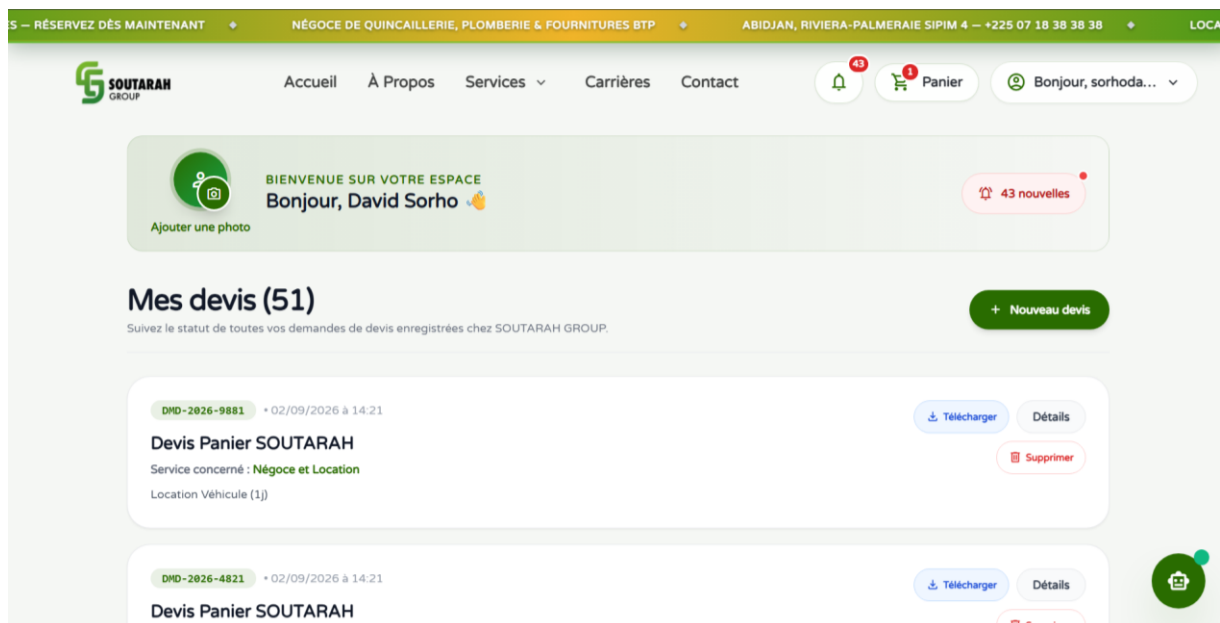


Figure 26 : Web Interface : Customer Portal and Request Tracking

IV. IMPLEMENTATION OF THE MOBILE RESERVATION APPLICATION

The 'SOUTARAH Mobile' application provides a tailored response to mobility by focusing on the vehicle rental service:

1. Home Screen and Detailed Vehicle Sheet

The home screen displays available vehicles categorized (SUVs, Sedans, Commercial) with dynamic filters, daily rental rates, and high-definition visuals. The detailed vehicle sheet presents vehicle technical specifications (air conditioning, transmission, fuel type), incorporates a calendar selector, and dynamically computes total cost based on destination and driver option.

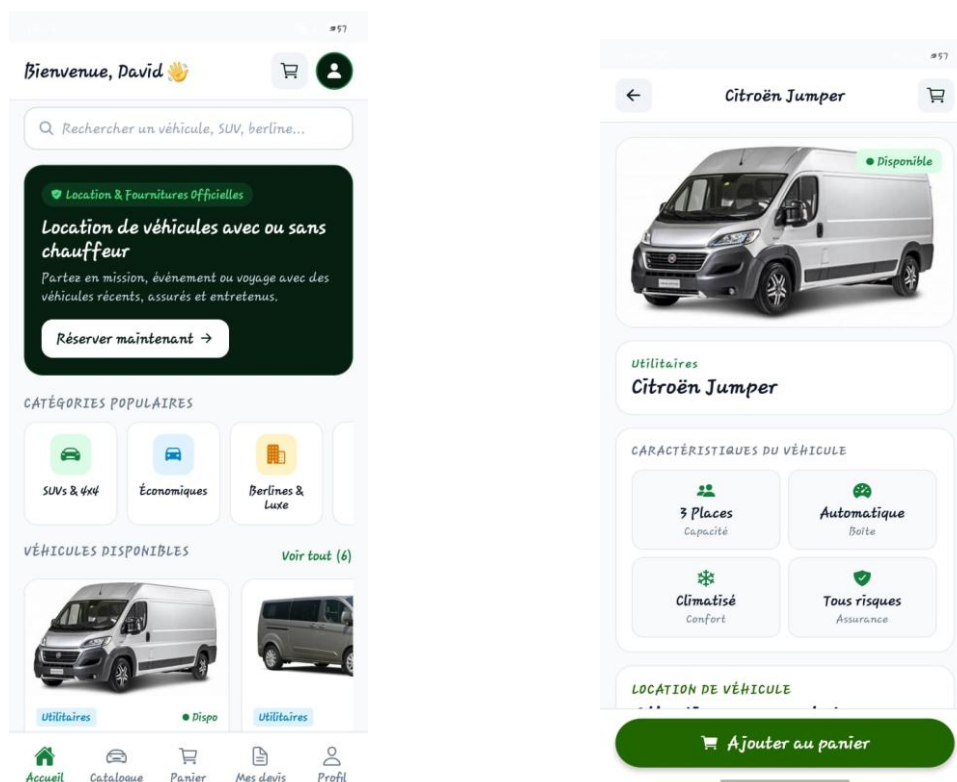


Figure 27 : Mobile Application : Home Screen and Detailed Vehicle Sheet

2. Mobile Booking Cart

Allows adjusting rental quantities or durations, providing specific delivery instructions, and confirming requests in real time.

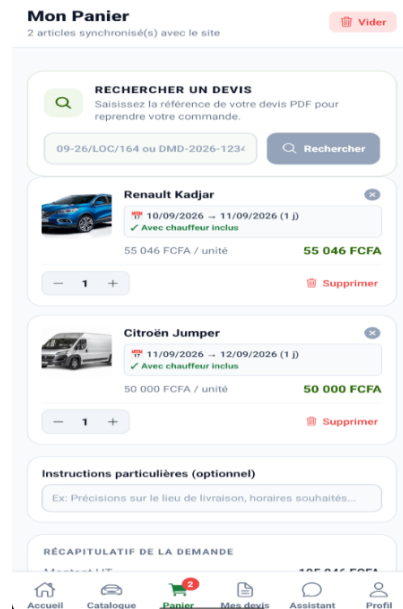


Figure 28 : Mobile Application : Mobile Booking Cart and Validation

V. IMPLEMENTATION OF THE ADMINISTRATOR PORTAL

The Web Back-Office serves as the enterprise control tower. It includes:

- Dashboard: performance charts, projected revenue, active quotation counts, and current reservations;
- Fleet and Trading Inventory Management: adding new vehicles, products, and updating daily rates;

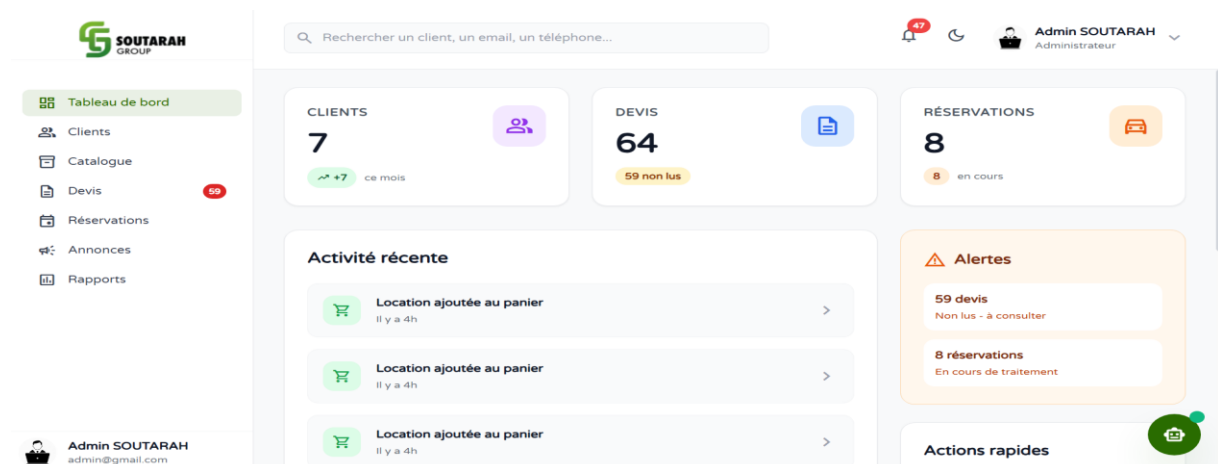
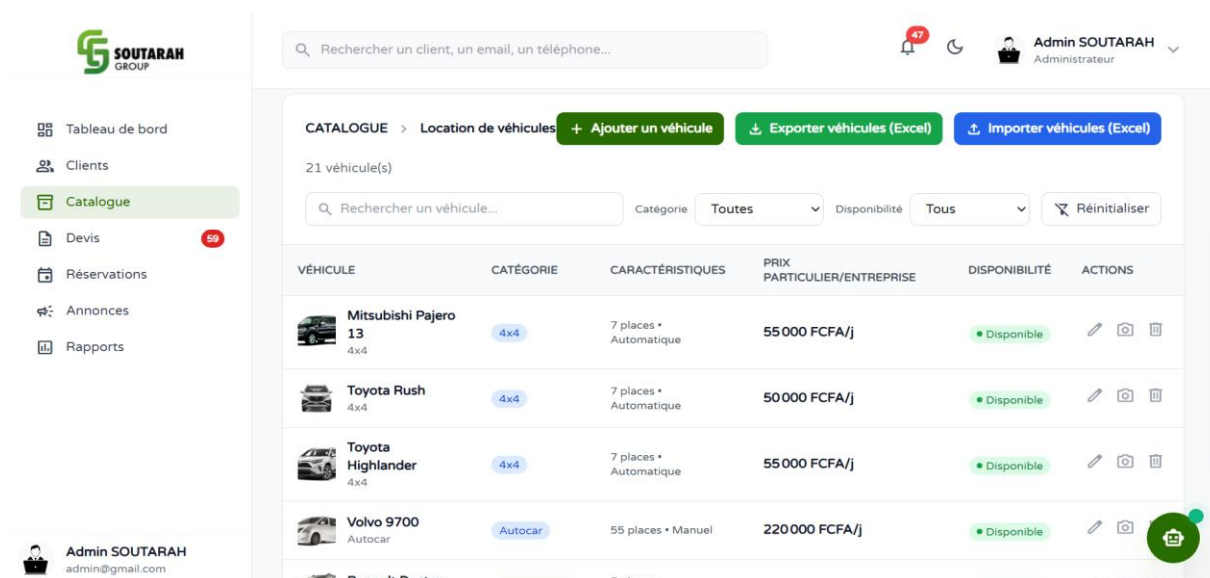


Figure 29 : Administrative Interface : Dashboard



CATALOGUE > Location de véhicules + Ajouter un véhicule Exporter véhicules (Excel) Importer véhicules (Excel)

21 véhicule(s)

Rechercher un véhicule... Catégorie Toutes Disponibilité Tous Réinitialiser

VÉHICULE	CATÉGORIE	CARACTÉRISTIQUES	PRIX PARTICULIER/ENTREPRISE	DISPONIBILITÉ	ACTIONS
 Mitsubishi Pajero 13 4x4	4x4	7 places • Automatique	55 000 FCFA/j	Disponible	  
 Toyota Rush 4x4	4x4	7 places • Automatique	50 000 FCFA/j	Disponible	  
 Toyota Highlander 4x4	4x4	7 places • Automatique	55 000 FCFA/j	Disponible	  
 Volvo 9700 Autocar	Autocar	55 places • Manuel	220 000 FCFA/j	Disponible	  
 Renault Duster		5 places •			  

Admin SOUTARAH
admin@gmail.com

Figure 30 : Administrative Interface : Fleet and Reservation Management

CHAPTER 6

SYSTEM DEPLOYMENT

I. FUNCTIONAL TESTING AND VALIDATION

An exhaustive testing campaign was conducted to validate the robustness and integrity of the interconnected solution:

Module / Feature	Executed Test Scenario	Expected Result	Status
Authentication	User login with valid email and password	JWT token allocation and session opening	COMPLIANT
API Security	Attempt to access administration route without privileges	Request blocked and HTTP 403 Forbidden response	COMPLIANT
Web Catalog	Browsing and filtering multi-division products	Instant display of matching items	COMPLIANT
Mobile Reservation	Selecting a vehicle and choosing rental dates	Exact dynamic price calculation (duration x price + driver option)	COMPLIANT
Availability Check	Attempt to reserve during an already booked period	Date conflict alert and submission blocked	COMPLIANT
PDF Quotation Generation	Cart validation and PDF document download	Automatic generation of standard 2-page PDF quotation	COMPLIANT

DB Synchronization	Mobile order validation and Back-Office verification	Immediate appearance of order in admin list	COMPLIANT
Email Notifications	Transmission of a new quotation request	Automatic reception of alert email via Brevo	COMPLIANT

Table 4 : Test Acceptance Book and Functional Validation (Web & Mobile)

II. FINANCIAL ASSESSMENT AND COST ESTIMATION

1. Hardware and Software Costs

Since all selected development technologies (React, React Native, Node.js, MySQL, Expo) are based on free Open Source licenses, direct software development licensing costs are zero. Hardware investments were focused on the development workstation and mobile test devices.

2. Hosting and Maintenance Costs

For production deployment, estimated annual operating costs are broken down as follows:

Expense Item	Technical Description	Estimated Annual Cost (FCFA)
Domain Name & SSL	.com domain name + SSL certificate	35,000
Mobile Developer Accounts	Google Play Store (one-time) + Apple Developer (annual)	75,000
Preventive Maintenance	Automated backups, security updates and technical support	180,000
TOTAL ESTIMATED ANNUAL	Overall operating and maintenance budget	290,000 FCFA

Table 5 : Estimated Financial Summary of Infrastructure and Annual Maintenance Costs

III. Profitability, Return on Investment (ROI) and Perspectives

1. Immediate Profitability and Return on Investment

Automating the booking process and quote generation allows SOUTARAH GROUP to reduce administrative processing time per client file by more than 80% (dropping from 45 minutes to under 3 minutes). This immediate commercial responsiveness, combined with the total elimination of operating losses related to double bookings, yields a full return on investment starting from the very first quarter of operations.

2. Platform Evolution Perspectives

To support the company's growth, the modular architecture established will enable deploying in the short and medium terms:

- **Loyalty Program** : Awarding reward points for each reservation or order, redeemable for discounts or free services, to enhance customer retention;
- **Telematics and GPS Tracking** : Integration of connected trackers on vehicles to monitor trips live and schedule preventive mechanical maintenance;
- **Native Push Notifications** : Instant smartphone alerts via Firebase Cloud Messaging for reservation updates;

GENERAL CONCLUSION

At the conclusion of this graduation internship carried out within the company SOUTARAH GROUP, we successfully completed the comprehensive design and realization of a unified digital solution, addressing the theme: « **Design and development of a digital platform for managing the services of SOUTARAH GROUP integrating a mobile application dedicated to vehicle reservation** ».

This project profoundly transformed the commercial operations of the company by replacing manual practices with a modern, consistent, and interconnected software ecosystem. The rigorous methodological approach based on the Unified Process and UML modeling (UP/UML) guaranteed precise requirement analysis and a flawless relational schema for the MySQL database.

From a technical standpoint, the synergy between the web platform developed with React.js and the mobile application engineered with React Native/Expo, both orchestrated around a Node.js/Express REST API, achieved all stated objectives. Users now benefit from complete visibility across the company's 6 business divisions, a seamless reservation flow with an anti-double-booking algorithm, an interactive multi-item shopping cart, and instant generation of official PDF quotations.

On a personal and academic level, this project proved to be an exceptionally enriching experience. It allowed us to consolidate our theoretical foundation acquired at the École Supérieure d'Industrie (ESI) of INP-HB, grasp real-world software engineering constraints within an enterprise environment, and master cutting-edge full-stack technologies.

BIBLIOGRAPHY

I. REFERENCE BOOKS AND METHODOLOGICAL MANUALS :

- AUDIBERT, Laurent. « UML Course: Object Modeling and Diagrams », Éditions Eyrolles / University Institute, pages 21-46.
- ROQUES, Pascal. « Practical UML 2: Case Studies and Corrected Exercises », Éditions Eyrolles, 7th edition, 2021, 394 pages.

II. ACADEMIC COURSE MATERIALS (INP-HB / ESI) :

- Zana Yéo. « HTTP-Server-and-Express.js », Lecture Notes, STIC, ESI / INP-HB Yamoussoukro, 2025-2026.

WEBOGRAPHY

- REACT.JS. « Official React 18 and Virtual DOM Documentation », Meta Open Source. Accessible at : <https://react.dev> (Consulted on August 30, 2026).
- REACT NATIVE & EXPO. « Cross-Platform Native Mobile Development with Expo Framework ». Accessible at : <https://reactnative.dev> and <https://docs.expo.dev> (Consulted on September 15, 2026).
- BREVO. Accessible at : <https://brevo.com> (Consulted on September 1, 2026).

TABLE OF CONTENTS

DEDICATION	- 1 -
ACKNOWLEDGEMENTS.....	- 2 -
FOREWORD	- 3 -
TABLE OF CONTENTS	- 5 -
LIST OF FIGURES.....	- 6 -
LIST OF TABLES	- 7 -
LIST OF ABBREVIATIONS	- 8 -
ABSTRACT.....	- 9 -
GENERAL INTRODUCTION	- 10 -
PART I : PROJECT FRAMEWORK AND CONTEXT	- 11 -
Chapter 1 : Overview of the Host Organization	- 12 -
I. Presentation of SOUTARAH GROUP	- 12 -
1. General Presentation	- 12 -
2. Mission, Vision and Values.....	- 12 -
3. Business Divisions and Services.....	- 13 -
4. Partnerships	- 14 -
Chapter 2 : Project Presentation	- 15 -
I. Project Context and Justification	- 15 -
III. Existing System Study and Limitations	- 15 -
III. Project Objectives	- 15 -
1. General Objective.....	- 15 -
2. Specific Objectives	- 16 -
IV. System Specifications	- 16 -
1. Functional Requirements on the Customer Side.....	- 16 -
2. Functional Requirements on the Administrator Side	- 17 -
3. Technical Requirements and Constraints	- 17 -
V. Task Planning and GANTT Diagram	- 17 -
1. Forecast Planning	- 17 -
2. GANTT Diagram	- 18 -
PART II : CONCEPTUAL STUDY.....	- 19 -
Chapter 3 : Choice of Analysis Methodology	- 20 -
I. Comparative Study of Analysis Methodologies	- 20 -
1. Overview of the MERISE Methodology	- 20 -
2. Overview of the UP/UML Methodology.....	- 20 -
3. Comparative Table : MERISE and UP/UML	- 21 -
4. Choice of Analysis Methodology.....	- 22 -
II. Application of UP/UML to the SOUTARAH GROUP System	- 22 -
1. Use Case Diagrams	- 23 -
❖ Textual Description of Use Cases.....	- 23 -
2. Activity Diagrams	- 27 -
2.1. Customer Account Creation and Management	- 27 -
2.2. Vehicle Reservation with Availability Check	- 28 -
2.3. Product Purchase Process.....	- 30 -
3. Sequence Diagrams.....	- 32 -
3.1. Secure Online Payment	- 32 -
3.2. Vehicle Reservation	- 33 -
3.3. Product Purchase	- 34 -
3.4. User Registration	- 35 -
4. Class Diagram	- 36 -
PART III : TECHNICAL STUDY.....	- 37 -
Chapter 4 : System Design and Technological Choices	- 38 -



I. Analysis and Justification of Technological Choices	- 38 -
1. Development Environment	- 38 -
2. UI/UX Wireframing Tool	- 38 -
3. Frontend Web Technologies	- 39 -
4. Mobile Technologies	- 39 -
5. Backend Technologies and REST API	- 40 -
6. Notification and Messaging Technologies	- 40 -
7. Hosting and Deployment Platform	- 40 -
II. Database Management System (DBMS)	- 41 -
1. Comparison of Major DBMS	- 41 -
2. Selection and Justification of MySQL	- 42 -
III. General System Architecture	- 42 -
PART IV : IMPLEMENTATION AND OPERATIONAL DEPLOYMENT	- 44 -
Chapter 5 : Development of the Web Platform and Mobile Application	- 45 -
I. MySQL Database Schema Implementation	- 45 -
II. Implementation of the AI Assistant and Notifications (Brevo)	- 46 -
1. Intelligent Virtual Assistant (AI Agent)	- 46 -
2. Messaging and Notification Service (Brevo / SMTP)	- 46 -
3. The secure online payment module (Genius Pay)	- 47 -
III. Development of Web Platform Interfaces	- 47 -
1. Home and Authentication Pages	- 47 -
2. Product Catalog	- 49 -
3. Vehicle Reservation Module	- 49 -
4. Multi-Item Cart and PDF Quotation Generation	- 50 -
5. Customer Portal and Request Tracking	- 50 -
IV. Development of the Mobile Reservation Application	- 51 -
1. Home Screen and Detailed Vehicle Sheet	- 51 -
2. Mobile Booking Cart	- 52 -
V. Development of the Administrator Space (Web Back-Office)	- 52 -
Chapter 6 : Implementation	- 54 -
I. Functional Testing and Validation (Acceptance Book)	- 54 -
II. Financial Assessment and Cost Estimation	- 55 -
1. Hardware and Software Costs	- 55 -
2. Annual Hosting and Maintenance Costs	- 55 -
III. Profitability, Return on Investment and Perspectives	- 56 -
1. Immediate Profitability and Return on Investment (ROI)	- 56 -
2. Platform Evolution Perspectives	- 56 -
GENERAL CONCLUSION AND PERSPECTIVES	- 57 -
BIBLIOGRAPHY	- 58 -
I. Academic Works and Courses	- 58 -
II. Technical Documentation and Web References	- 58 -
WEBOGRAPHY	- 59 -
TABLE OF CONTENTS	- 60 -